# Notes on Advanced Macroeconomics

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# Preface

Aim of this work of mine is to provide clear macroeconomic knowledges in language as simple as possible. Most of this note is taken from the advanced macroeconomics course of Yanfei Deng I took. This note is still a work in progress and needs tons of ammending. See the newest version at `https://github.com/xolarvill/notes-on-economics` along with my other notes. And pull a request if you figure some part of this note is wrong.
This work was first written in markdown language on Obsidian, which provides a lightweight and highly personalized working experience. But it was later transfered into purely latex language on Sublime Text, due to poor support for rendering of long math equation blocks of Obsidian. For the tranfer I used Pandoc at `https://pandoc.org/`. I wrote some LaTeX snippets of Sublime Text to optimize the writing experience, and uploaded on github at `https://github.com/xolarvill/snippets-for-quick-latex-on-st`.
One can find enormous academical help on advanced macroeconomics from books listed below:

- Introduction to Modern Economic Growth, D. Acemoglu
- Economic Growth, R. Barro and S.i.M. Xavi
- The ABCs of RBCs, An Introduction to Dynamic Macroeconomic Models, G. McCandless
- Macroeconomics, A Comprehensive Textbook for First-Year Ph.D. Courses in Macroeconomics. M. Azzimonti, P. Krusell, A. McKay, and T. Mukoyama
- Adavanced Macroeconomics, D. Romer

# 1 Introduction

These notes assume intermediate macroeconomics, including national accounts, basic growth theory, and aggregate demand and supply. We now ask how aggregate outcomes follow from decisions over time, and how those decisions respond to the environment and policy.

The first three parts cover growth, fluctuations, and the open economy. Part IV studies money, finance, and policy; Part V applies these mechanisms to the dollar system and China's growth after the pandemic. Chapters and sections marked *planned* contain outlines for later development.

## 1.1 Growth, fluctuations, and their interaction

Growth theory asks how productive capacity and living standards evolve. Starting with capital accumulation, we introduce saving decisions, population, technology, and resource allocation to explain differences across economies.

Business-cycle theory asks how an economy responds to disturbances in productivity, demand, or financing. We identify the shock, then explain the response of spending, employment, and prices. A mechanism that amplifies a shock need not be its cause.

The questions overlap. A recession can reduce investment and change future capital. Population and productivity growth can change the conditions for stabilization policy. Growth need not be an unaffected trend, and fluctuations can have lasting effects. We begin with separate models to isolate mechanisms, then study their interaction.

## 1.2 From aggregate relations to individual decisions

An aggregate consumption function such as $C = C(Y - T)$ summarizes a relationship. It does not yet explain how a household responds to an expected tax change, uncertain income, or a borrowing limit. To study those questions, we specify what the household chooses and what constrains it.

**A two-period example.** Let a household receive real income $y_0$ today and $y_1$ tomorrow, with no initial or terminal assets. Saving $a_1$ earns a known gross real return $R > 0$; negative $a_1$ is borrowing at that return. The problem is

$$\begin{aligned} \max_{c_0, c_1, a_1} \; & u(c_0) + \beta u(c_1), \qquad 0 < \beta < 1, \\ \text{s.t. } & c_0 + a_1 = y_0, \\ & c_1 = y_1 + R a_1. \end{aligned} \tag{1}$$

Assume $u' > 0$, $u'' < 0$, positive consumption, and an interior solution. Substituting the constraints leaves one choice:

$$\max_{a_1} \; u(y_0 - a_1) + \beta u(y_1 + R a_1). \tag{2}$$

Differentiate with respect to $a_1$, then substitute back for consumption:

$$-u'(y_0 - a_1) + \beta R u'(y_1 + R a_1) = 0, \tag{3}$$

$$u'(c_0) = \beta R u'(c_1). \tag{4}$$

Saving one more unit costs $u'(c_0)$ units of utility today and yields $\beta R u'(c_1)$ tomorrow. An interior optimum equates the two.

For marginal utility $u'(c) = c^{-\sigma}$, $\sigma > 0$,

$$c_0^{-\sigma} = \beta R c_1^{-\sigma}, \tag{5}$$

$$\left(\frac{c_1}{c_0}\right)^{\sigma} = \beta R, \tag{6}$$

$$\frac{c_1}{c_0} = (\beta R)^{1/\sigma}. \tag{7}$$

Thus consumption growth depends on patience and the return to saving. Its sensitivity to that return is governed by $1/\sigma$, the elasticity of intertemporal substitution in this example. A binding borrowing constraint would require an inequality condition instead of this interior equality.

**Expectations and policy.** With uncertainty, today's utility cost is known but tomorrow's gain is not. Replace that gain by its conditional expectation. For an asset with gross real payoff $R_{t+1}$, the interior condition becomes

$$u'(c_t) = \beta E_t[R_{t+1} u'(c_{t+1})], \qquad E_t[\cdot] \equiv E[\cdot \mid \mathscr{I}_t]. \tag{8}$$

Here $\mathscr{I}_t$ is information available when the household chooses saving. An uncertain return stays inside the expectation; its comovement with marginal utility matters. Rational expectations require forecasts consistent with the model and available information, not knowledge of tomorrow's realization.

The same logic matters for policy. An announced future tax change may alter consumption today. A change in a policy rule can change expectations and the aggregate relationships observed under the old rule. This is the concern in the Lucas critique.[1] Writing down individual decisions makes these responses explicit. It does not, by itself, guarantee that the chosen preferences, constraints, or information assumptions remain appropriate under every policy change.

## 1.3 Reading a dynamic equilibrium

The household takes income and returns as given. In general equilibrium, they must be consistent with production and other agents' choices. A competitive equilibrium combines individual optimality, budget constraints, market clearing, and policy rules at each date and, under uncertainty, each history.

**States, choices, and paths.** Consider a deterministic economy with fixed labor normalized to one, output $f(k_t)$, and no government or foreign trade. Capital $k_t$ is available at the start of period $t$. With depreciation $\delta$, feasibility gives

$$k_{t+1} = f(k_t) + (1 - \delta)k_t - c_t. \tag{9}$$

Capital is inherited; consumption is chosen and affects tomorrow's capital. Here $k_t$ is a state variable and $c_t$ a control variable. The set of states depends on the model's timing and assumptions.

If households can save in capital, competitive factor pricing gives the gross return $f'(k_{t+1}) + 1 - \delta$. Substituting this return into (4) yields the intertemporal condition

$$u'(c_t) = \beta\left[f'(k_{t+1}) + 1 - \delta\right] u'(c_{t+1}). \tag{10}$$

Together, (9) and (10) describe feasible accumulation and optimal consumption growth. Given $k_0$, different initial choices of $c_0$ generate different candidate paths. Feasibility and the Euler equation alone do not establish that each path is an equilibrium. The model also needs its boundary conditions, including the

[1] Robert E. Lucas, Jr. (1976), "Econometric Policy Evaluation: A Critique," *Carnegie-Rochester Conference Series on Public Policy*, 1, 19–46. https://doi.org/10.1016/S0167-2231(76)80003-6.

appropriate transversality condition for this infinite-horizon saving problem. We derive these conditions in the Ramsey chapter.

**A steady state is a particular path.** For a positive stationary allocation, set $k_{t+1} = k_t = k^*$ and $c_{t+1} = c_t = c^*$ in the two equations. Since $u'(c^*) > 0$,

$$c^* = f(k^*) - \delta k^*, \tag{11}$$

$$1 = \beta \left[ f'(k^*) + 1 - \delta \right]. \tag{12}$$

These equations locate a steady state. They do not describe the transition from an arbitrary $k_0$, or prove existence or uniqueness of an equilibrium path. With sustained technological or population growth, aggregate quantities may keep growing on a balanced growth path. A stationary system may then require normalization, for example by effective labor, rather than constant levels.

**Equilibrium and welfare answer different questions.** A planner chooses a feasible allocation directly; a competitive equilibrium also specifies prices that support individual choices. Under suitable conditions, a planner's allocation can be decentralized. Taxes, market power, externalities, incomplete markets, or nominal rigidities may prevent that equivalence. Market clearing alone establishes neither efficiency nor the desirability of a policy. A welfare comparison must state whose utility is evaluated, which constraints apply, and what alternative is feasible.

## 1.4 Accounting and notation used in these notes

We need only a few accounting conventions at the outset. In a closed economy, using gross investment $I_t$,

$$Y_t = C_t + I_t + G_t, \tag{13}$$

$$K_{t+1} = (1 - \delta)K_t + I_t. \tag{14}$$

$K_t$ is the stock available at the beginning of the period; $I_t$ is a flow during the period. Depreciation already enters (14). We do not add it again to (13). A household's purchase of a financial claim need not create physical capital; domestic claims and liabilities must be consolidated when deriving aggregate feasibility.

Accounting becomes a model only after we specify behavior, technology, and equilibrium. Government budgets enter with fiscal policy; international accounts enter in Part III.

Capital letters generally denote aggregates; lower-case letters denote per-person or normalized quantities. Check each chapter's normalization before comparing equations. Unless stated otherwise, $\beta$ is a discount factor, $\rho$ a rate of time preference, $r$ a net real interest rate, and $R = 1 + r$ a gross real return. In discrete time, the convention $\beta = 1/(1 + \rho)$ differs from continuous-time discounting $e^{-\rho t}$.

For a nominal one-period return $1 + i_t$, realized inflation $\pi_{t+1} = P_{t+1}/P_t - 1$ gives the gross real return

$$R_{t+1} = \frac{1 + i_t}{1 + \pi_{t+1}}. \tag{15}$$

An interest rate quoted in nominal units is therefore not the real return in (8). Under uncertainty, the realized price change must remain inside the relevant expectation.

Finally, for a positive variable $X_t$ and a positive stationary reference $\bar{X}$, define

$$\hat{x}_t \equiv \log X_t - \log \bar{X} \approx \frac{X_t - \bar{X}}{\bar{X}}. \tag{16}$$

The definition is exact; the approximation is local. A deviation from a steady state, a growth rate, and an output gap relative to a flexible-price allocation are different objects. Each chapter states its reference point before using an approximation.

# Part I
# Growth

How does an economy's productive capacity evolve? We begin with growth facts, then move from an imposed saving rate to optimal saving, overlapping generations, and endogenous technological change.

# 2 Growth Facts and Growth Accounting

*Planned chapter.*

## 2.1 Income levels, growth rates, and convergence

Distinguish differences in income levels from differences in growth rates. Set out the convergence patterns that the growth models need to explain.

## 2.2 Capital, labor, and productivity

Derive a growth-accounting decomposition and state its measurement assumptions. Explain why an accounting contribution is not, by itself, a causal explanation.

## 2.3 From accounting to a growth model

Use the decomposition to motivate capital accumulation, saving decisions, and technology as separate objects of analysis.

# 3 Solow Model

The Solow-Swan model is an economic model of long-run growth, focusing on capital accumulation, population growth, and technological progress. Developed by Robert Solow and Trevor Swan in 1956, it replaced the Keynesian Harrod-Domar model.

> Mathematically, the Solow-Swan model is a nonlinear system consisting of a single ordinary differential equation that models the evolution of the per capita stock of capital.

Learning points for 23-2 Course:

- the economic environment of Solow model
- the main equations in Solow model
- Several key concepts of Solow model, i.e. golden rate, equilibrium dynamics
- how does the model converge under different conditions

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## 3.1 Economic Environment and Specific Assumptions

Solow model seems to have simple assumptions, but in which many things are pre-assumed.
**Sections**

- A Solow model assumes only two sections in participation

$$Y(t) = C(t) + I(t) = C(t) + S(t) \tag{17}$$

**Household**

- A large number of non-optimizing homogeneous households (with ad-hoc assumptions) $\Rightarrow$ representative agent
- Constant saving rate of $s \in [0, 1] \Rightarrow S(t) = sY(t) = I(t)$

**Firm**

- A large number of non-optimizing homogeneous firms (with ad-hoc assumptions) $\Rightarrow$ representative agent
- Since sharing a common production method, it can use an aggregate production function

**Market structure**

- Competitive $\Rightarrow$ Solow model is a prototypical competitive general equilibrium model

**Endowment (labor and capital)**

- Labor $L(t)$ provided inelastically by households at wage $w(t)$
- Households also own the capital $K(t)$ and rent it to firms at capital rate $R(t)$

- The inital capital is given as $K(0)$
- Capital exponential depreciation is denoted as $\delta$ $\Rightarrow$ real interest rate $r(t) = R(t) - \delta$
- How the capital is distributed is irrelevant to our exploring

**Market clearing**

- Labor market clearing condition is $L(t) = \bar{L}(t)$
- Capital market clearing condition is $K^s(t) = K^d(t)$

**Technology**

- Technology is free and publicly available

**Production function**
The aggregate production function is

- continuous
- differentiable
- positive
- diminishing MP
- CRTS

**Production function in Harrold form**

$$Y_t = F(K_t, A_t L_t) \tag{18}$$

- This format of production function means, in the final, the ratio of capital to production $K/Y$ will be stable.
- It is convient to make $A_t$ times $L_t$ instead of other ways in Solow Model.

> **Note 1** (Production function in Hicks neutrality form)**.**
>
> $$Y_t = A_t F(K_t, L_t)$$

**CRTS**

$$F_t(cK_t, cL_t) = cF(K_t, L_t),\ \forall c \geqslant 0 \tag{19}$$

- One way to see CRTS as a reasonable requirement is that imagining the scale of economy large enough to cover the potiential benefits of cooperation.
- Another way is imagining all other factors are compared irrelevant than capital, labor and knowledge.
- The CRTS assumption can derive the production function to $y = f(k)$.

**Inada Conditions**

$$\lim_{x \to 0} y = \lim_{x \to 0} f(x) = \infty \tag{20}$$

$$\lim_{x \to \infty} y = \lim_{x \to \infty} f(x) = 0 \tag{21}$$

- Many have argued that Inada conditions are the key limits of Solow model, as well as the exogenosu tech growth rate.
- The purpose of this setting is to confine the model from wild spreading.
- Cobb-Douglas Function is very suitable in this setting. *It is rather reducted than deducted from the economic nature.*

**Note 2** (No profits for firms)**.**

$$\Rightarrow \textit{PMP:} \tag{22}$$

$$\max_{L(t),K(t)} \; F - w(t)L(t) - R(t)K(t) \tag{23}$$

$$\Rightarrow \begin{cases} w(t) = F_L \\ R(t) = F_K \end{cases} \tag{24}$$

$$\Rightarrow Y(t) = w(t)L(t) + R(t)K(t) \textit{ (applying Euler's equation)} \tag{25}$$

$$\textit{Meaning the firms make no profits. They are just profit-maximizing.} \tag{26}$$

**Main equations**
All of these above should be considered oversimplication of the real world, but to the favor of an economist.

$$\begin{cases} y_t = F(K_t, A_t N_t) \\ I_t = S_t = sY_t \\ K_{t+1} = I_t + (1-\delta)K_t \end{cases} \tag{27}$$

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## 3.2 Solow model in discrete time

**Note 3** (Impulse response function)**.** *Impulse response functions are useful for studying the interactions between variables in a vector autoregressive model. They represent the reactions of the variables to shocks hitting the system. It is often not clear, however, which shocks are relevant for studying specific economic problems.*

**Note 4.** *Even the edge or limit of a dynamic state is no way near the steady state.*

**Fundamental law of motion**

$$\text{Solow model's motion equations} \begin{cases} Y_t = F(K_t, A_t, L_t) \\ K_{t+1} = (1-\delta)K_t + I_t \end{cases} \tag{28}$$

**Question 1** (how to solve this set of difference equations?)**.** *See Gao Xu, 2017 and McCandless, 2008*

$$y_t = \frac{Y_t}{N_t} = A_t \cdot F(\frac{K_t}{N_t}, \frac{N_t}{N_t}) \equiv A_t f(k_t) \tag{29}$$

$$K_{t+1} = (1-\delta)K_t + I_t \tag{30}$$

$$\Rightarrow \frac{K_{t+1}}{N_{t+1}} \frac{N_{t+1}}{N_t} = \frac{(1-\delta)K_t}{N_t} + \frac{I_t}{N_t} \tag{31}$$

$$\Rightarrow k_{t+1} = \frac{(1-\delta)k_t + i_t}{1+n} \tag{32}$$

$$\Rightarrow (1+n)k_{t+1} = (1-\delta)k_t + i_t \text{ (transpose)} \tag{33}$$

$$\Rightarrow (1+n)k_{t+1} = (1-\delta)k_t + s_t \tag{34}$$

$$\Rightarrow (1+n)k_{t+1} = (1-\delta)k_t + s \cdot (1+\alpha)^t A_0 \cdot y_t \text{ ($\alpha$ is the growth rate for tech, in simplest case= 0)} \tag{35}$$

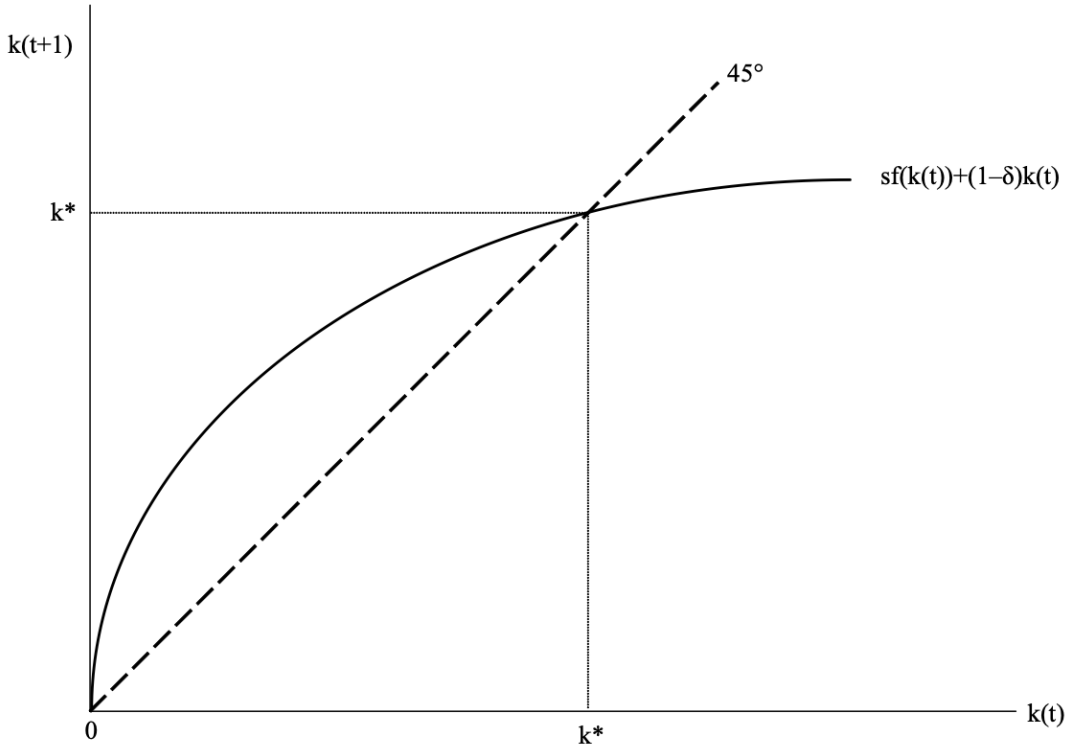
$$\Rightarrow (1+n)k_{t+1} = (1-\delta)k_t + sA_0 y_t \tag{36}$$

It is obvious the steady state of capital per worker can be found at when $k_{t+1} = k_t = k^*$

$$k_{t+1} = g(k_t) = \frac{(1-\delta)k_t + sA_0 f(k_t)}{1+n} \tag{37}$$

**Equilibrium steady states in a phase diagram**

With $A_t$ and $N_t$ being stable (not have to be constant), the solution of difference equations in discrete Solow model is $k_{t+1} = sf(k_t) + (1-\delta)k_t$, which leads to the phase diagram below




**Equilibrium dynamics, solving the $k^*$**

$$k_{t+1} = sf(k_t) + (1-\delta)k_t \tag{38}$$

$$= g(k_t) \tag{39}$$

$$\Rightarrow \text{eventually that } k^* = g(k^*) \tag{40}$$

$$\Rightarrow \frac{k^*}{f(k^*)} = \frac{s}{n+\delta} \text{ (meaning capital-output ratio is anchored by the parameters)} \tag{41}$$

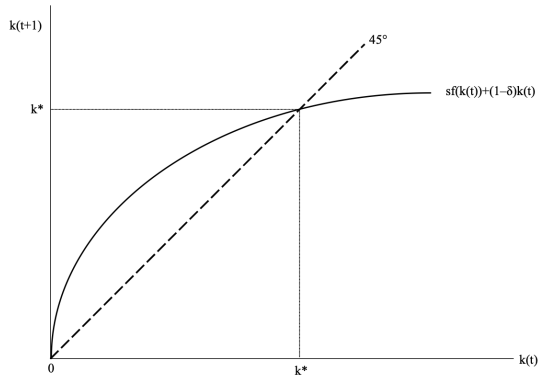
$$\Rightarrow k^* = \cdot \text{ (solving the real final } k) \tag{42}$$

**Growth rate**

The growth rate of capital is denoted as

$$\gamma_t = \frac{k_{t+1}}{k_t} = \frac{(1-\delta)k_t + sA_0 f(k_t)}{(1+n)k_t} \tag{43}$$

**Intertemporal equilibrium, convergence speed and convergence time**




$$k_{t+1} = g(k_t) \tag{44}$$
$$\Rightarrow k_{t+1} = k^* + g'(k^*)(k_t - k^*) \tag{45}$$
$$\Rightarrow k_{t+1} = k^* + g'(k)(k_t - k^*) \tag{46}$$
$$\Rightarrow \hat{k_{t+1}} \equiv \frac{k_{t+1} - k^*}{k^*} = g'(k^*)(\frac{k_t - k^*}{k^*}) = g'(k^*)\hat{k_t} \text{ (redefining)} \tag{47}$$
$$\Rightarrow \hat{k_t} = [g'(k^*)]^t \hat{k_0} \text{ (by the method of iteration, the equil steady state of capital growth ratio)} \tag{48}$$
$$\Rightarrow t = \frac{\log \frac{\hat{k_t}}{\hat{k_0}}}{\log g'(k^*)} \text{ (using logarithms)} \tag{49}$$
$$\Rightarrow t = \frac{\log \frac{k_t - k^*}{k_0 - k^*}}{\log g'(k^*)} \text{ (meaning the time } t \text{ is now decided by the location of capital } k_t) \tag{50}$$

We have the convergence speed $g'(k^*)$ and convergence time $t$.

**Note 5** (Steady State vs Intertemporal Equilibrium)**.** *Despite of seemingly different division, the two are sides of a coin.*

**Balance of growth**

$$k_t = \frac{K_t}{A_t L_t} \tag{51}$$
$$\Rightarrow \frac{dK_t}{k_t} = \frac{dk_t}{k_t} + \frac{dA_t}{A_t} + \frac{dN_t}{N_t} \tag{52}$$

**Convergence under different conditions:**

i) first scenerio: $\begin{cases} A_t = a \\ N_t = n \end{cases}$

$$k_t \equiv \frac{K_t}{N} \tag{53}$$

$$y_t \equiv \frac{Y_t}{N} = F[\frac{K_t}{N}, A] = f(k_t) \tag{54}$$

$$k_{t+1} = sf(k_t) + (1-\delta)k_t \tag{55}$$

$$\text{The steady state equilibrium is } k(t) = k^*, \ \forall t \tag{56}$$

ii) second scenerio: $\begin{cases} A_t = a \\ \frac{N_t}{N_{t-1}} = 1 + n \end{cases}$

iii) third scenerio: $\begin{cases} \frac{A_t}{A_{t-1}} = 1 + a \\ \frac{N_t}{N_{t-1}} = 1 + n \end{cases}$

---

## 3.3 Solow model in continuous time

Let $A(t) = 1$, we have the Solow model in continous time

$$Y(t) = F[K(t), L(t)] \tag{57}$$

$$I(t) = S(t) = sY(t) \tag{58}$$

$$\dot{K(t)} = I_t - \delta K_t \tag{59}$$

Derivation of the basic function of Solow model in continuous time, which is the steady state of Solow model

$$k(t) = \frac{K(t)}{L(t)} \tag{60}$$

$$\frac{\dot{k(t)}}{k(t)} = \frac{\dot{K(t)}}{K(t)} - \frac{\dot{L(t)}}{L(t)} \tag{61}$$

$$\frac{\dot{k(t)}}{k(t)} = \frac{sF - \delta K(t)}{K(t)} - n \tag{62}$$

$$\frac{\dot{k(t)}}{k(t)} = \frac{(sF - \delta K(t)/L(t))}{K(t)/L(t)} - n \tag{63}$$

$$\Rightarrow \dot{k(t)} = sf[k(t)] - (\delta + n)k(t) \tag{64}$$

**Intertemporal equilibrium, convergence speed and convergence time**

$$\dot{k(t)} = sf[k(t)] - (\delta + n)k(t) \equiv g[k(t)] \tag{65}$$

$$\Rightarrow \text{eventually at steady state } g(k^*) = 0 = sf[k^*] - (\delta + n)k^* \tag{66}$$

$$\Rightarrow \frac{k^*}{f(k^*)} = \frac{s}{\delta + n} \text{ (the anchored capital-output ratio determined by parameters)} \tag{67}$$

$$\Rightarrow \dot{k(t)} \approx g(k^*) + g'(k^*)[k(t) - k^*] \text{ (linearization by using Taylor expansion)} \tag{68}$$

$$\Rightarrow \frac{dK(t)}{dt} = g'(k^*)k(t) - g'(k^*)k^* \tag{69}$$

$$\Rightarrow k(t) = [k(0) - k^*]e^{[g'(k^*)t]} + k^* \tag{70}$$

$$\Rightarrow \text{transposition} \tag{71}$$

$$\text{Denote } \hat{k(t)} \equiv \frac{k(t) - k^*}{k^*} = e^{[g'(k^*)t]}\frac{k(0) - k^*}{k^*} \tag{72}$$

$$\Rightarrow \frac{\hat{k(t)}}{\hat{k(0)}} = e^{g'(k^*)t} \tag{73}$$

$$\Rightarrow \text{Denote } t = \frac{log[\frac{\hat{k(t)}}{\hat{k(0)}}]}{g'(k^*)} \tag{74}$$

$$\Rightarrow \text{Like in discrete time, we have the convergence speed } g'(k^*) \text{ and time for convergence } t \tag{75}$$

---

## 3.4 Golden Rate

One of the philosophies upon which economic science is established is utilitism. In approximate understanding, that is achieving max happiness by max consuming. To do so in a stationary state in Solow model, the idea of golden rate is introduced.
Golden rate in discrete time, or the maximized welfare of Solow model

$$\text{Starting with two ad-hoc assumptions } \begin{cases} C_t = (1 - s)Y_t \\ S_t = sY_t \end{cases} \tag{76}$$

$$\Rightarrow \text{At the steady state:} \tag{77}$$

$$c^*(s) = (1 - s)f[k^*(s)] = f[(1 - s)k^*(s)] = f[k^*(s)] - (n + \delta)k^*(s) \tag{78}$$

$$\Rightarrow \max c^* \tag{79}$$

$$\Rightarrow s_g = \underset{s}{\text{argmax}}\ c^* \text{ (golden rate of saving)} \tag{80}$$

$$\Rightarrow k_g^* = \underset{k^*}{\text{argmax}}\ c^* \text{ (golden rate of capital per capita)} \tag{81}$$

**Question 2** (Why does Solow model matter?)**.**

Jess Benhabib and Mark M. Spiegel, 1994, The role of human capital in economic development: Evidence from aggregate cross-country data. Journal of Monetary Economics 34: 143-17

> *"Our results indicate that human capital enters insignificantly in explaining per capita growth rates."*

Lant Pritchett, 2001, Where has all the education gone? The World Bank Economic Review 15(3): 367-391.

> *"Cross-national data show no association between increases in human capital attributable to the rising educational attainment of the labor force and the rate of growth of output per worker. This implies that the association of educational capital growth with conventional measures of total factor production is large, strongly statistically significant, and negative."*

Chandra Shekhar Kumar, 2006, Human capital and growth empirics. The Journal of Developing Areas 40(1): 153-179.

> *"In all the above estimation methodologies, the positive and significant effect of human capital is not observed."*

There must be various reasons to explain the fact. Pritchett examines three possible reasons. But, before arguing the reasons, one inquiry seems to be missing. It is the question of validity of Solow augmented production function and by consequence the validity of endogenous growth theory (at least that based on the Lucas-Uzawa model). Three papers above cited imply that there is no significant positive effect of human capital accumulation on the increase of per capita product.

## 3.5 Extensions

Solow model with money

- Purvis, Douglas D. "Introducing Useful Money into a Growth Model." The Canadian Journal of Economics / Revue Canadienne d'Economique 4, no. 3 (1971)

Augmented Solow model

- N. Gregory Mankiw, David Romer, David N. Weil, A Contribution to the Empirics of Economic Growth, The Quarterly Journal of Economics, Volume 107, Issue 2, May 1992, Pages 407–437

AK model (often treated as a tool to introduce endogeneous growth model)

- Romer, Paul M. (1986). "Increasing Returns and Long-Run Growth". Journal of Political Economy. 94 (5): 1002–1037.

Malthus to Solow (unified growth theory)

- Hansen, Gary, D., and Edward C. Prescott. 2002. "Malthus to Solow ." American Economic Review, 92 (4): 1205–1217.

# 4 Ramsey Model

See Chapter 5 to Chapter 7 from Acemoglu, 2004 for more mathematical help.

We begin the road to sophiscated macro skyscraper with the most basic frame, the Ramsey model.
The Ramsey model[2], or the RCK model, or the Neoclassical growth model differs from the Solow model in that the choice of consumption is explicitly microfounded at a point in time and so endogenizes the savings rate. Ramsey model is in nature a growth model with consumer optmization.
The most intuitive change is now we put $S_t$ under $\max$ as $\max_{\{C_t, S_t\}}$.

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## 4.1 Assumptions

- Same as Solow Model
  - Production Function in the Harrold neutrality form
  - **Exogenous Variables:**
  - Population growth rate= $n$
  - Technology growth rate= $g$
  - Depreciation rate= $0$ (for simplicity)
  - And most of all, the technological growth rate is assumed as $a$.
- Firm
  - Perfect competitive firms, or homogenous firms, or representive firms
  - Hire workers and capital in competitive factor markets
  - Each has the production function $Y = F(K, AL)$
  - Firms are owned by the families, meaning the revenue goes straight to latter
- Household
  - Homogenous families.
  - Population growth of each household is assumed as $n$.
  - Each member in a household provides one unit of labor.
  - Income of a household is the sum of the labor revenue subsum, the capital revenue subsum and the firm revenue share subsum.
  - A household should and must maximize the utility.
  - Degree of patience denoted as discount factor $\beta = \frac{1}{1+\rho}$, and $\rho$ is the discount rate.

**Note 6** (Rule of Prohibiting Ponzi Schemes)**.** *Conconstraint in Ramsey model is also called Prohibiting Ponzi Schemes Rule. It forbids the household to do a scam of rolling over debt by repeatedly borrowing. Yet in real life, many people are doing this as a harmless way to ralex debt pressure.*

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[2]Ramsey, Frank P. (1928). "A Mathematical Theory of Saving". Economic Journal. 38 (152): 543–559.

## 4.2 Solution of Ramsey model

### 4.2.1 Central Planner's Problem

Originally Ramsey set out the model as a social planner's problem of maximizing levels of consumption over successive generations. Only later was a model adopted by Cass and Koopmans as a description of a decentralized dynamic economy with a representative agent. This is also why the model is called Ramsey-Cass-Koopsman model, yet abbreviated as Ramsey model.
So, chronologically the first thing we should address is central planner's problem.
**Disperssive competition**
UMP

$$\begin{aligned} &\max_{\{C_t,K_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(C_t) \\ &s.t. C_t + K_{t+1} - (1-\delta)K_t = W_t L_t + R_t k_t + D_t \end{aligned} \tag{82}$$

Noticing that the constraint is equivalent to $C_t + I_t = Y_t + D_t$
Solving this UMP resulting Euler's equation

$$\begin{aligned} &\Rightarrow u'[F(K_t) - K_{t+1}] = u'(C_t) = \beta u'(C_{t+1})F'(K_{t+1}) \\ &\iff \frac{u'(C_t)}{u'(C_{t+1})} = \beta R_{t+1} \end{aligned} \tag{83}$$

PMP at a perfectly competitive market

$$\max_{K_t} \ \pi = F(K_t, A_t L_t) - (r+\delta)K_t - wL_t \tag{84}$$

where the $A_t L_t$ is assumed as a constant in order for model to converge

$$F.O.C. \Rightarrow r + \delta = R = 1 + r \tag{85}$$

In a conclusion, optimization in the disperssive economy yields $\begin{cases} \frac{u'(C_t)}{u'(C_{t+1})} = \beta R_{t+1} \\ r + \delta = R = 1 + r \end{cases}$

**Central Planner's Problem**

$$\begin{aligned} &\max_{\{C_t,K_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(C_t) \\ &s.t.\ C_t + K_{t+1} = Y_t, \text{ where } \delta \text{ is assumed as 1} \end{aligned} \tag{86}$$

Using Lagranian method and acquire Euler's equation

$$\begin{aligned} &\Rightarrow u'(F(K_t) - K_{t+1}) = u'(C_t) = \beta u'(C_{t+1})F'(K+1) \\ &\iff \frac{u'(C_t)}{u'(C_{t+1})} = \beta R_{t+1} \end{aligned} \tag{87}$$

**Conclusion**
Regardless of the economy being disperssive or central planning, we would have the same Euler's equation of consumption. This suggests for simplicity, we might as well use central planner's problem.

### 4.2.2 Method of Lagrangian

We start from scratch, then move on to methods like using Hamiltonian equation or using Bellman equation.
**one-period static optimization**

$$\begin{aligned} &v = f(x_1, x_2) \\ &s.t.\ g(x_1, x_2) = z \end{aligned} \tag{88}$$

**two-period dynamic optimization**
A general form is denoted as

$$\begin{aligned} &\max v = f(x_1, x_2) \\ &s.t.\ g(x_1, x_2) = z \text{ (where the subscript is denoted as time periods)} \end{aligned} \tag{89}$$

a two-period consumer maximization problem can be denoted as

$$\begin{aligned} &\underset{C_1, C_2}{max}\ u_1 = u(C_1) + \beta u(C_2) \\ &s.t. \begin{cases} C_1 + S_1 \leqslant [r + (1-\delta)]S_0 + Y_1 \\ C_2 + S_2 \leqslant [r + (1-\delta)]S_1 + Y_2 \end{cases} \text{,where we assume } \begin{cases} S_0 = S_2 = 0 \\ C_1, C_2 > 0 \end{cases} \end{aligned} \tag{90}$$

$$\text{The reconstraints} \Rightarrow C_1 + \frac{C_2}{1+r} = Y_1 + \frac{Y_2}{1+r} \text{ (intertemporal budget conconstraint)} \tag{91}$$

**Question 3** (How to solve the intertemporal UMP?)**.**

(1). method of elimination
$\frac{du_1}{dC_1} = 0 \Rightarrow \frac{\beta u'(C_2)}{u'(C_1)} = \frac{1}{1+r}$ ( the Euler's equation of intertemporal consumption) In which the $\frac{1}{1+r}$ is the comparative price
(2). method of geomotry
On a $C_1 - C_2$ plane,

$$0 = u'(C_1)dC_1 + \beta u'(C_2)dC_2 \tag{92}$$

$$\Rightarrow \frac{dC_2}{dC_1} = -\frac{u'(C_1)}{\beta u'(C_2)} \tag{93}$$

(3). method of Lagrangian function

$$\mathscr{L} = u(C_1) + \beta u(C_2) + \lambda[C_1 + \frac{C_2}{1+r} - Y_1 - \frac{Y_2}{1+r}] \tag{94}$$

$$\Rightarrow \begin{cases} \frac{\partial L}{\partial C_1} = 0 \\ \frac{\partial L}{\partial C_2} = 0 \\ \frac{\partial L}{\partial \lambda} = 0 \end{cases} \tag{95}$$

$$\Rightarrow \frac{\beta u'(C_2)}{u'(C_1)} = \frac{1}{1+r} \text{ (in the difference equation lies the core of Ramsey model )} \tag{96}$$

**t-period dynamic optimization (finite-period)**
1) The original version:

$$\begin{aligned} &\max_{C_1,C_2,...,C(T)} \; u_1 = u(C_1) + \beta u(C_2) + \beta^2 u(C_3) + ... + \beta^{T-1} u(C_T) \\ &s.t. \begin{cases} C_1 + S_1 & \leqslant [1+r]S_0 + Y_1 \\ C_2 + S_2 & \leqslant [1+r]S_1 + Y_2 \\ & \dots \\ C_T + s_T & \leqslant (1+r)S_{T-1} + Y_T \end{cases} \text{,where we assume} \begin{cases} \delta = s_0 = s_T = 0 \\ C_1, C_2, ..., C_T > 0 \end{cases} \end{aligned} \tag{97}$$

where the budget constraints can be condensed into

$$C_1 + \frac{C_2}{1+r} + ... + \frac{C_T}{(1+r)^{T-1}} = Y_1 + \frac{Y_2}{1+r} + ... + \frac{Y_T}{(1+r)^{T-1}} \tag{98}$$

2) Or the second version:

$$\begin{aligned} &\max_{\{C_t\}_{t=1}^{\infty}} \; u_1 = \sum_{t=1}^{T} \beta^{t-1} u(C_t) \\ &s.t. \begin{cases} C_1 + S_1 & \leqslant [1+r]S_0 + Y_1 \\ C_2 + S_2 & \leqslant [1+r]S_1 + Y_2 \\ & \dots \\ C_T + S_T & \leqslant (1+r)S_{T-1} + Y_T \end{cases} \text{,where we assume} \begin{cases} S_0 = S_T = 0, \delta = 0 \\ C_1, C_2, ..., C_T > 0 \end{cases} \end{aligned} \tag{99}$$

To solve this t-period UMP, using Langranian method: (the $\lambda$ is the shadow price)

$$\max_{\{C_t\}_{t=1}^{\infty}, \lambda} \; \mathscr{L} = \sum_{t=1}^{T} \beta^{t-1} u(C_t) + \lambda [c_1 + \frac{c_2}{1+r} + ... + \frac{C_T}{(1+r)^{T-1}} - (Y_1 + \frac{Y_2}{1+r} + ... + \frac{Y_T}{(1+r)^{T-1}})] \tag{100}$$

$$\Rightarrow F.O.C.s \begin{cases} \frac{\partial \mathscr{L}}{\partial c_1} = 0 \\ \frac{\partial \mathscr{L}}{\partial C_2} = 0 \\ \vdots \\ \frac{\partial \mathscr{L}}{\partial C_T} = 0 \end{cases} \tag{101}$$

$$\Rightarrow \frac{\beta u'(C_{t+1})}{u'(C_t)} = \frac{1}{1+r}, t = 1, 2 \dots, T \tag{102}$$

3) Another commonly adopted simplified version is

$$\begin{aligned} &\max_{\{C_t, S_t\}_{t=1}^{T}} \; u_1 = \sum_{t=1}^{T} \beta^{T-1} u(C_t) \\ &s.t. C_t + S_t \leqslant (1+r)S_{t-1} + Y_t, \text{ where} \begin{cases} C_t > 0 \\ S_0 = S_T = 0 \end{cases} \end{aligned} \tag{103}$$

Using Lagrangian function method:

$$\Rightarrow u'(C_{t+1}) = \lambda_{t+1} \tag{104}$$

$$\Rightarrow \frac{u'(C_{t+1})}{u'(C_t)} = \frac{\lambda_{t+1}}{\lambda_t} = \frac{1}{\beta(1+r)} \tag{105}$$

$$\Rightarrow \frac{\beta u'(C_{t+1})}{u'(C_t)} = \frac{1}{1+r} \tag{106}$$

**infinite-period dynamic optimization**
Using the third version, we delve into infinity

$$\begin{aligned} &\max_{\{C_t\}_{t=1}^{\infty}} u_1 = \sum_{t=1}^{\infty} \beta^{T-1} u(C_t) \\ &s.t. C_t + S_t \leqslant (1+r) S_{t-1} + Y_t, \text{ where } \begin{cases} C_t > 0 \\ S_0 = \lim\limits_{t\to\infty} S_T = 0 \end{cases} \end{aligned} \tag{107}$$

Using Lagrangian function method:

$$\mathscr{L} \equiv \sum_{t=0}^{\infty} \{\beta^t u(c_t) + \lambda_t [(1+r) s_{t-1} + Y_t - (c_t + s_t)]\} \tag{108}$$

$$\Rightarrow F.O.C. \tag{109}$$

$$\Rightarrow u'(c_{t+1}) = \lambda_{t+1} \tag{110}$$

$$\Rightarrow \frac{u'(c_{t+1})}{u'(c_t)} = \frac{\lambda_{t+1}}{\lambda_t} = \frac{1}{\beta(1+r)} \tag{111}$$

$$\Rightarrow \frac{\beta u'(c_{t+1})}{u'(t)} = \frac{1}{1+r} \tag{112}$$

### 4.2.3 Method of Hamiltonian Equation (Optimal Control)

Method of Hamiltonian equation is part of the optimal control theory from mathematicals.
**a) discrete time**

$$\begin{aligned} &\max_{\{c_t, s_t\}_{t=0}^{T}} u_1 = \sum_{t=1}^{T} \beta^{T-1} u(c_t) \\ &s.t. c_t + s_t \leqslant (1+r) s_{t-1} + Y_t, \text{ where } \begin{cases} c_t > 0 \\ s_{-1} = s_T = 0 \\ \beta = \frac{1}{1+\rho} \end{cases} \end{aligned} \tag{113}$$

**b) continuous time without patience**

$$\begin{aligned} &\max_{\{C(t), S(t)\}} u = \int_{t=0}^{T} e^{-\rho t} u[C(t)] dt = \int_{t=0}^{T} u[C(t)] dt \\ &s.t.\ C(t) + \dot{S}(t) = r \cdot S(t) + Y(t), \text{ where } S(0) = S(T) = 0 \end{aligned} \tag{114}$$

Using Lagrangian

$$\Rightarrow \mathscr{L} = \int_{t=0}^{T} \{u[C(t)] + \lambda(t)[C(t) + \dot{S}(t) - r \cdot S(t) - Y(t)]\} dt \tag{115}$$

$$= \int_{t=0}^{T} \{H(t) - \lambda(t)\dot{S}(t)\} dt \ (H \text{ is Hamiltonian}) \tag{116}$$

$$= \int_{t=0}^{T} H(t)dt - \int_{t=0}^{T} \lambda(t)\dot{S}(t)dt \ (\text{Then applying fractional integral method to latter}) \tag{117}$$

$$= \int_{t=0}^{T} H(t)dt - \{\lambda(t)S(t)|_0^T - \int_{t=0}^{T} \dot{\lambda}(t)S(t)dt\} \tag{118}$$

$$= \int_{t=0}^{T} H(t)dt - \dot{\lambda}(t)S(t)dt - [\lambda(t)S(t)]|_0^T \tag{119}$$

$$= \int_{t=0}^{T} H(t)dt + \dot{\lambda}(t)dt \tag{120}$$

with this simplified version containing Hamiltonian

$$\Rightarrow \text{F.O.C. of Lagrangian function: } \begin{cases} \frac{\partial \mathscr{L}}{\partial C(t)} = 0 \ (C(t) \text{ is the control variable}) \\ \frac{\partial \mathscr{L}}{\partial S(t)} = 0 \ (S(t) \text{ is the state variable}) \\ \frac{\partial \mathscr{L}}{\partial \lambda(t)} = 0 \ (\lambda(t) \text{ is the monodromy variable}) \end{cases} \tag{121}$$

$$\Rightarrow \text{the optimal control } \begin{cases} \frac{\partial \mathscr{H}}{\partial C(t)} = 0 \\ \frac{\partial \mathscr{H}}{\partial S(t)} = -\dot{\lambda}(t) \\ \frac{\partial \mathscr{H}}{\partial \lambda(t)} = \dot{S}(t) \end{cases} \tag{122}$$

**Note 7** (control variable and state variable)**.** *If a variable is determined in the current period, it is a control variable. If it is determined period(s) before, it is a state variable.*

**c) continuous time with patience**

Next we consider the Lagrangian function with subjective discount rate

Right into this version's Lagrangian function

$$\mathscr{L} = \int_{t=0}^{T} \{e^{-\rho t}u[C(t)] + \lambda(t)[C(t) + \dot{S}(t) - r \cdot S(t) - Y(t)]\} dt \tag{123}$$

$$= \int_{t=0}^{T} \{e^{-\rho t}u[C(t)] + \lambda(t)[rS(t) + Y(t) - C(t)] - \lambda\dot{S}(t)\} dt \tag{124}$$

$$= \int_{t=0}^{T} \{H(0) - \lambda\dot{S}(t)\} dt \ (H(0) \text{ is present value Hamiltonian}) \tag{125}$$

$$\Rightarrow H(0) \equiv e^{-\rho t}u[C(t)] + \lambda(t)[rS(t) + Y(t) - C(t)] \tag{126}$$

$$\Rightarrow \tilde{H}(t) = e^{\rho t}H(0) \text{ (the current value Hamiltonian)} \tag{127}$$

$$\Rightarrow H(t) = u[C(t)] + \eta(t)[rS(t) + Y(t) - C(t)] \tag{128}$$

Comparing the two scenarios (with and without consideration for $\rho$), we can see — regardless of the subjective discount rate, result would be the same.

### 4.2.4 Method of Bellman Equation (Dynamic Programming)

**a) discrete time**
UMP as below

$$\begin{aligned} &\max_{\{C_t,S_t\}_{t=0}^{\infty}} u_0 = \beta^t u(C_t) \\ &s.t.\ C_t + S_t = (1+r)S_{t-1} + Y_t, \text{ where } S_1 = \lim_{t\to\infty} s_t = 0 \end{aligned} \tag{129}$$

Using the budget constraint to rewrite utility function

$$\Rightarrow C_t = (1+r)S_{t-1} + Y_t - S_t \tag{130}$$

$$\Rightarrow u_0 = \max_{\{S_t\}_{t=0}^{\infty}} \beta^t u\big[(1+r)S_{t-1} + Y_t - S_t\big] \tag{131}$$

$$= \max_{S_0} \beta^t u\big[(1+r)S_{-1} + Y_0 - S_0\big] + \beta \max_{\{S_t\}_{t=1}^{\infty}} \sum_{t=1}^{\infty} \beta^{t-1} u\big[(1+r)S_{t-1} + Y_t - S_t\big] \tag{132}$$

$$= \max_{s_0} u\big[(1+r)S_{-1} + Y_0 - S_0\big] + \beta u(S_0, Y_1) \tag{133}$$

Upscript $o$ denotes optimal

$$\Rightarrow u^o(S_{t-1}, Y_t) = \max_{S_t}\{u\big[(1+r)S_{t-1} + Y_t - S_t\big] + \beta u^o(S_t, Y_{t+1})\} \text{ (Bellman equation)} \tag{134}$$

$$= \max_{S_t}\{u(C_t) + \beta u^o(S_t, Y_{t+1})\} \tag{135}$$

$$\Rightarrow \text{F.O.C.} : u_{C_t}\frac{\partial C_t}{\partial S_t} + \beta\frac{du(S_t, Y_{t+1})}{dS_t} = 0 \tag{136}$$

Let $\frac{\partial C_t}{\partial S_t}$ be $-1$

$$\Rightarrow \beta\frac{du(s_t, Y_{t+1})}{ds_t} = u_{c_t} \tag{137}$$

Move one period ahead and acquire Euler's equation

$$\tag{138}$$

$$u(s_t, Y_{t+1}) = \max_{s_{t+1}}\{u\big[(1+r)s_t + Y_t - s_{t+1}\big] + \beta u(s_{t+1}, Y_{t+2})\} \tag{139}$$

$$\Rightarrow \frac{du^o(s_t, Y_{t+1})}{ds_t} = \frac{\partial u(\cdot)}{\partial s_t} + \frac{\partial u(\cdot)}{\partial s_{t+1}}\frac{ds_{t+1}}{ds_t} + \beta\frac{du^o(\cdot)}{ds_t}\frac{ds_{t+1}}{ds_t} \tag{140}$$

$$= \frac{\partial u(\cdot)}{\partial s_t} + \big[\frac{\partial u(\cdot)}{\partial s_{t+1}} + \beta\frac{du^o(\cdot)}{d_{s_{t+1}}}\big]\frac{ds_{t+1}}{ds_t} \text{ (using optimals' envelope therome )} \tag{141}$$

$$= u_{c,t+1}\frac{dc_{t+1}}{ds_t} + 0 \tag{142}$$

$$= u_{c,t+1}\cdot(1+r) \tag{143}$$

$$\Rightarrow \beta\frac{u_{c,t+1}}{u_{c,t}} = \frac{1}{1+r} \tag{144}$$

Hence the same result as before.
**b) continuous time (Hamilton-Jacob-Bellman, HJB)**
See in ...

## 4.3 Intertemporal Equilibrium of Ramsey model

### 4.3.1 Dynamics of c

Recalling that

$$\frac{\dot{c(t)}}{c(t)} = \frac{r(t) - \rho - \theta g}{\theta} \tag{145}$$

Since we know

$$r(t) = f'[k(t)] \tag{146}$$

Combining them as

$$\frac{\dot{c(t)}}{c(t)} = \frac{r(t) - \rho - \theta g}{\theta} = \frac{f'[k(t)] - \rho - \theta g}{\theta} \tag{147}$$

Once we assume that

$$\exists k^* \in \mathbb{R}, f'[k^*] = \rho - \theta g \tag{148}$$

The dynamics of $c$, denoted as $\frac{\dot{c(t)}}{c(t)}$ is purely dependent on the value of $k$ according to Ramsey model.

### 4.3.2 Dynamics of k

Since in Ramsey model we don't assume depreciation of capital $k$, denoted as $\delta = 0$. The dynamics of $k$ accordingly is $\dot{k}(t) = f[k(t)] - c(t) - (n+g)k(t)$.
This way, in order for the dynamics of $k$ returning to 0, the model must fulfil the requirement of $c(t) = f[k(t)] - (n+g)k(t)$.

### 4.3.3 Phase diagram and dynamic efficiency

Combining the the two dynamics together, we would have

$$\dot{k} = 0 \tag{149}$$

$$\dot{c} = 0 \tag{150}$$

in a coordinate system of $k$ and $c$.

**Note 8** (dynamic efficiency in Ramsey model)**.** *Beware that the equilibrium of two dynamics would always be on the left of golden rate of $k$. The origin point $(0,0)$ can also be considered a special case of equilibrium.*

The equilibrium in Ramsey model has given only one equilibrium path.

Recalling the Solow presented as model equation of motion

$$k_{t+1} - (1-\delta)k_t = sF(k_t) \tag{151}$$

RCK model has two equations of motion

$$\begin{cases} \frac{u'(c_t)}{u'(c_{t+1})} = \beta[F'(k_{t+1}) - (1-\delta)] \\ k_{t+1} - (1 - \delta k_t) = F(k_t) - C_t \end{cases} \tag{152}$$

Acquirring the steady state of Ramsey model

$$\begin{cases} k_{t+1} = k_t = k^*, \triangle k = 0 \\ c_{t+1} = c_t = c^*, \triangle c = 0 \end{cases} \tag{153}$$

$$\Rightarrow F'(k^*) = \rho + \delta \Rightarrow k^* \tag{154}$$

$$\Rightarrow c^* = F(k^*) - \delta k^* \tag{155}$$

$$\Rightarrow (k^*, c^*) \text{ is the steady state} \tag{156}$$

$$\Rightarrow \frac{dc^*}{dk^*} = F'(k) - \delta = R - \delta \tag{157}$$

Deriving the derivatives at steady state

$$\begin{cases} \frac{\partial \triangle c}{\partial k} < 0 \\ \frac{\partial \triangle k}{\partial c} < 0 \end{cases} \tag{158}$$

, meaning there must be seperating line for each

$$\triangle c = 0 \tag{159}$$

$$\Rightarrow F'(k^*) = \rho + \delta \text{ (from the first equation of motion of RCK)} \tag{160}$$

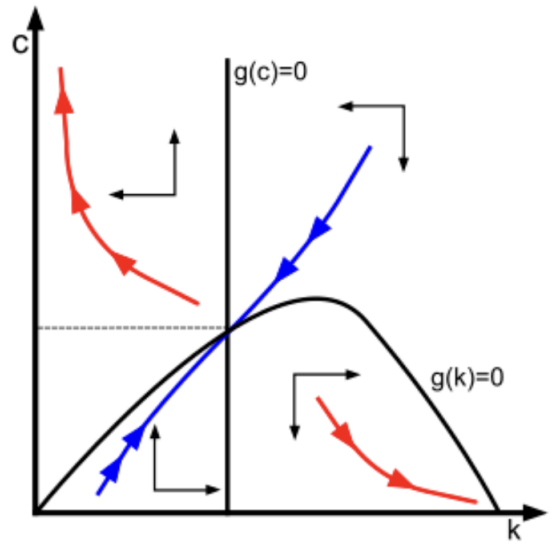
$$\Rightarrow F_{kk} dk = 0 dc \tag{161}$$

$$\Rightarrow \frac{dc}{dk} = \frac{F_{kk}}{0} \Rightarrow \text{the seperating line is vertical in C-K plane} \tag{162}$$

$$\triangle k = 0 \tag{163}$$

$$\Rightarrow C^* = F(k^*) - \delta k^* \tag{164}$$

$$\Rightarrow \frac{dc}{dk} = F_k - \delta \Rightarrow \text{the seperating line is vertically reversed U shape} \tag{165}$$




## 4.4 Linearization

Comparing Solow and Ramsey:
Solow model's Euler equation

$$\dot{k}(t) + \delta k(t) = sF(k(t)) \tag{166}$$

Ramsey model's Euler equations

$$\dot{c}(t) = \frac{F'[k(t) - (\delta + \rho)]}{\theta} c(t), \text{ where } \theta = -\frac{cu''(c)}{u'(c)} \tag{167}$$

$$\dot{k}(t) = F[k(t)] - \delta k(t) - c(t) \tag{168}$$

Noticing the two equations of Ramsey are unlinear. This demands a process of linearization.
**Linearization, method of Taylor expansion**
First method we use the same technique from Solow model's linearization, Taylor's Expansion.
First order Taylor expansion of RCK' Euler equations:

$$\begin{bmatrix} \dot{c}(t) \\ \dot{k}(t) \end{bmatrix} = \begin{bmatrix} \frac{F'[k(t)-(\delta+\rho)]}{\theta} c(t) \\ F[k(t)] - \delta k(t) - c(t) \end{bmatrix} = \begin{bmatrix} \phi^1[c(t), k(t)] \\ \phi^2[c(t), k(t)] \end{bmatrix} \tag{169}$$

$$\Rightarrow \begin{bmatrix} \dot{c}(t) \\ \dot{k}(t) \end{bmatrix} = \begin{bmatrix} \phi^1[c^*, k^*] \\ \phi^2[c^*, k^*] \end{bmatrix} + \begin{bmatrix} \phi^1_c & \phi^1_k \\ \phi^2_c & \phi^2_k \end{bmatrix} |_{c^*,k^*} \begin{bmatrix} c(t) - c^* \\ k(t) - k^* \end{bmatrix} \tag{170}$$

$$= \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} o & \frac{F''(k^*)}{\theta} c^* \\ -1 & \rho \end{bmatrix} \begin{bmatrix} c(t) - c^* \\ k(t) - k^* \end{bmatrix} \tag{171}$$

The above deduction use the following conclusions

$$\begin{cases} F'(k^*) - (\delta + \rho) = 0 \Rightarrow F'(k^*) = \rho + \delta \Rightarrow k^* \\ c^* = F(k^*) - \delta k^* \Rightarrow c^* \end{cases} \tag{172}$$

Because ...

$$Ax = \lambda x \tag{173}$$
$$\Rightarrow \lambda^2 - tr(A)\lambda + dt + \lambda = 0 \tag{174}$$
$$\Rightarrow \lambda^2 - \rho\lambda + \frac{F'(k^*)}{\theta} \cdot c^* = 0 \tag{175}$$
$$\Rightarrow \lambda_1 = \frac{\rho - \sqrt{\rho^2 - 4F''(k^*)c^*}}{\theta} < 0 < \lambda_2 = \frac{\rho + \sqrt{\rho^2 - 4F''(k^*)c^*}}{\theta} \tag{176}$$

$\Rightarrow$ We choose the negative one which is $\lambda_1$ so that $k(t)$ can converge over time

$$\Rightarrow \dot{k}(t) = \lambda_1[k(t) - k^*] \tag{177}$$
$$\Rightarrow k(t) = k^* + e^{\lambda_1 t}[k(0) - k^*] \tag{178}$$

**Linearization, method of undetermined coefficients**
The second approach is method of undetermined coefficients

$$\dot{c}(t) = \frac{F''(k^*)}{\theta}c^* = 0 \tag{179}$$
$$\dot{k}(t) = \rho[k(t) - k^*] - [c(t) - c^*] \tag{180}$$
$$\Rightarrow \ddot{k}(t) = \rho\dot{k}(t) - \dot{c}(t) \tag{181}$$
$$= \rho\dot{k}(t) - \frac{F''(k^*)}{\theta} \cdot c^*[k(t) - k^*] \tag{182}$$

We can make an assumption that $\dot{k}(t) = \lambda[k(t) - k^*]$, which is a differential equation in nature

$$\Rightarrow \ddot{k}(t) = \lambda\dot{k}(t) \tag{183}$$
$$= \lambda^2[k(t) - k^*] \tag{184}$$
$$= \rho\dot{k}(t) - \frac{F''(k^*)c^*[k(t) - k^*]}{\theta} \tag{185}$$
$$= \lambda^2 - \rho\lambda + \frac{F''(k^*)c^*[k(t) - k^*]}{\theta} = 0 \tag{186}$$
$$\Rightarrow \lambda_1 = \frac{\rho - \sqrt{\rho^2 - 4F''(k^*)c^*}}{\theta} < 0 < \lambda_2 = \frac{\rho + \sqrt{\rho^2 - 4F''(k^*)c^*}}{\theta} \tag{187}$$

Samewise

$$\Rightarrow k(t) = k^* + e^{\lambda_1 t}[k(0) - k^*] \tag{188}$$

**Note 9** (convergence in Ramsey model)**.** *Recall Solow model's chapter of convergence speed and time, we can compare the two similar results:*

$$\begin{cases} \textit{Ramsey: } k(t) = k^* + e^{\lambda_1 t}[k(0) - k^*] \\ \textit{Solow: } k(t) = k^* + e^{[g'(k^*)t]}[k(0) - k^*] \end{cases} \tag{189}$$

*Meaning the $\lambda$ we are trying to derive here is the convergence speed.*

---

## 4.5 Maximizing welfare

Similiar to the concept of golden rate in Solow model, Ramsey model also provide a glimpse of maximized consumption.
At the steady state

$$c^* = F(k^*) - \delta k^* \tag{190}$$

$$\Rightarrow k^*_{gold} = \underset{k^*}{\text{argmax}}\ c^* \tag{191}$$

## 4.6 Taxation and capital accumulation

*Planned section.* Compare lump-sum and distortionary taxes in the neoclassical growth framework. Separate effects on steady-state capital from effects along the transition.

# 5 OLG Model

In contrast to the Ramsey model in which individuals are infinitely-lived, in the OLG model individuals live a finite length of time, long enough to overlap with at least one period of another agent's life.

**Note 10** (Meet only once)**.** *Two generation OLG is particularly simple because it precludes intertemporal trade (no one meets twice).*

**Solow model, RCK model and OLG model**

- All of them are under the assumption of perfect competitive market.
- Solow model has exogenous saving rate, whereas Ramsey model and OLG model has endogenous one.
- OLG model has heterogeneous representative agents.

---

## 5.1 Assumptions

$$C_t = L_t C_{1t}^{\text{young}} + L_{t-1} C_{2t}^{\text{old}} \text{ (where subscript 1 and 2 stands for age)} \tag{192}$$

$$L_t = (1+n)L_{t-1} \Rightarrow L_t = (1+n)^t L_0 \tag{193}$$

$$S_t = s_t L(t) \tag{194}$$

---

## 5.2 UMP of household

A general UMP would be

$$\max u(C_1) + \beta u(C_{2,t+1}) \quad s.t. \begin{cases} C_{1t} + S_t \leqslant W_t \\ C_{2,t+1} \leqslant R_{t+1} S_t \end{cases} \tag{195}$$

Form a Lagrangian and acquire Euler's equation

$$\mathscr{L} = u(C_1) + \beta u(C_{2,t+1}) + \lambda_{1t}(c_{1t} + s_t - w_t) + \lambda_{2,t+1}(c_{2,t+1} - R_{t+1} s_t) \tag{196}$$

$$\Rightarrow \text{F.O.C.} \begin{cases} \frac{\partial \mathscr{L}}{\partial C_{1t}} = 0 \\ \frac{\partial \mathscr{L}}{\partial C_{2,t+1}} = 0 \\ \frac{\partial \mathscr{L}}{\partial S_t} = 0 \end{cases} \Rightarrow \begin{cases} u'(C_{1t}) = \lambda_{1t} \\ \beta' u'(C_{2,t+1}) = \lambda_{2,t+1} \\ \frac{\lambda_{1t}}{\lambda_{2,t+1}} = R_{t+1} \end{cases} \tag{197}$$

$$\Rightarrow u'(C_{1t}) = \beta R_{t+1} u'(C_{2,t+1}) \text{ (Euler's equation in OLG UMP)} \tag{198}$$

e.g. Given a specific utillity function

$$u(C_{gt}) = \frac{C_{gt}^{1-\theta} - 1}{1-\theta}, 0 < \theta \neq 1, g = 1, 2 \tag{199}$$

$$\text{UMP} \Rightarrow \begin{cases} C_{1t}^{-\theta} = \beta R_{t+1} C_{2,t+1}^{-\theta} \\ C_{1t} = W_t - S_t \\ C_{2,t+1} = R_{t+1} S_t \end{cases} \tag{200}$$

$$\Rightarrow (W_t - S_t)^{-\theta} = \beta R_{t+1} (R_{t+1} S_t)^{-\theta} \tag{201}$$

$$\Rightarrow (\frac{W_t - S_t}{S_t})^{-\theta} = \beta R_{t+1}^{1-\theta} \tag{202}$$

$$\Rightarrow \frac{W_t}{S_t} = (\beta R_{t+1}^{1-\theta})^{-\frac{1}{\theta}} + 1 \tag{203}$$

$$\Rightarrow S_t = \frac{w_t}{\beta^{-\frac{1}{\theta}} R_{t+1}^{-\frac{1-\theta}{\theta}} + 1} < w_t \tag{204}$$

$$\Rightarrow S'_w = \frac{1}{\beta^{-\frac{1}{\theta}} R_{t+1}^{-\frac{1-\theta}{\theta}} + 1} \in (0, 1) \tag{205}$$

$$\Rightarrow S'_R = \frac{1-\theta}{\theta} \frac{S_t}{\beta^{-\frac{1}{\theta}} R_{t+1}^{-\frac{1-\theta}{\theta}} + 1} (\beta R_{t+1})^{-\frac{1}{\theta}} \tag{206}$$

$$\Rightarrow S'_R \begin{cases} > 0, 0 < \theta < 1 \\ = 0, \theta = 1 \\ < 0, \theta > 1 \end{cases} \tag{207}$$

---

## 5.3 PMP of firm

A general PMP would be

$$\max_{k_t, L_t} \pi_t = Y_t - w_t L_t - R_t K_t \tag{208}$$

$$\text{Since } \pi_t = \frac{\Pi_t}{L_t}, y_t = \frac{Y_t}{L_t}, k_t = \frac{K_t}{L_t}, \frac{K_{t+1}}{L_t} = \frac{K_{t+1}}{L_{t+1}} \frac{L_{t+1}}{L_t}, s_t = \frac{S_t}{L_t} \tag{209}$$

Rearrange in form of per capita

$$\Rightarrow \max_{K_t} \pi_t = Y_t - W_t - R_t K_t \tag{210}$$

$$\Rightarrow \text{F.O.C.} \begin{cases} R_t = f'(K_t) \\ W_t = f(K_t) - K_t f'(K_t) \end{cases} \tag{211}$$

---

## 5.4 Dynamics

Looking at the motion equation of capital

$$K_{t+1} - (1-\delta)K_t = I_t = S_t = s_t L_t = s(w_t, R_{t+1})L_t \tag{212}$$

$$\Rightarrow K_{t+1} = \frac{s(w_t, R_{t+1})}{1+n} \tag{213}$$

$$\Rightarrow K_{t+1} = \frac{w_t}{(\beta^{-\frac{1}{\theta}} R_{t+1}^{-\frac{1-\theta}{\theta}} + 1)(1+n)} \text{ (using conditions from UMP and PMP)} \tag{214}$$

$$\Rightarrow K_{t+1} = \frac{f(K_t) - K_t f'(K_t)}{(\beta^{-\frac{1}{\theta}} R_{t+1}^{-\frac{1-\theta}{\theta}} + 1)(1+n)} \text{ (equation of motion of capital in OLG)} \tag{215}$$

The steady states (intertemporal equilibrium) would be

$$k_{t+1} = k_t = k^* \tag{216}$$

$$\Rightarrow k^* = \frac{s[f(k_t^*) - k_t^* f'(k_t^*), f'(k^*)]}{1+n} \tag{217}$$

$$= \frac{f(k_t^*) - k_t^* f'(k_t^*)}{(\beta^{-\frac{1}{\theta}} f'(*)^{-\frac{1-\theta}{\theta}} + 1)(1+n)} \tag{218}$$

$$\Rightarrow k_{t+1} = \frac{(1-\alpha)(k^*)^\alpha}{(\beta^{-\frac{1}{\theta}}[\alpha(k^*)^{\alpha-1}]^{-\frac{1-\theta}{\theta}} + 1)(1+n)} \text{ (capital per capita at steady state in OLG)} \tag{219}$$

$$\Rightarrow k^* = \cdot \text{ (thus we have the } k \text{ at steady state)} \tag{220}$$

e.g., For a utillity function $u = \log c_{1t} + \beta \log c_{2,t+1}$, where $\theta = 1$.

$$\Rightarrow \frac{c_{2,t+1}}{c_{1t}} = \beta k_{t+1} \tag{221}$$

$$\Rightarrow s_t = \frac{w_t}{\beta^{-\frac{1}{\theta}} R_{t+1}^{-\frac{1-\theta}{\theta}} + 1} \tag{222}$$

$$= \frac{w_t}{\beta^{-1} + 1} \tag{223}$$

$$= \frac{\beta}{1+\beta} w_t \text{ (meaning the saving is a function of wage, instead of endowment)} \tag{224}$$

$$\Rightarrow k_{t+1} = \frac{s_t}{1+n} = \frac{\beta w_t}{(1+\beta)(1+n)} \tag{225}$$

$$= \frac{\beta(1-\alpha k_t^\alpha)}{(1+\beta)(1+n)} \tag{226}$$

$$\text{steady state demands } k_{t+1} = k_t = k^* \Rightarrow k^* = \left[\frac{\beta(1-\alpha)}{(1+\beta)(1+n)}\right]^{\frac{1}{1-\alpha}} \tag{227}$$

---

## 5.5 Comparing three models so far

$$\text{Solow}\begin{cases} L_t = (1+n)L_{t-1} \\ \frac{S_t}{L_t} = s_t = sy_t = s\frac{Y_t}{L_t} \\ k_{t+1} = \frac{(1-\delta)k_t + sf(k_t)}{1+n} \end{cases} \tag{228}$$

$$\text{OLG}\begin{cases} L_t = (1+n)L_{t-1} \\ k_{t+1} = \frac{s(w_t, R_{t+1})}{1+n} = \frac{s[f(k_t)-k_t f'(k_t), f'(k_{t+1})]}{1+n} \end{cases} \tag{229}$$

$$\text{Ramsey}\begin{cases} L_t = (1+n)L_{t-1} \\ k_{t+1} = \frac{f(k_t)+c_t}{1+n} \Rightarrow 0 = F(k^*) - c^* - \rho k^* \Rightarrow F'(k_G^*) = \delta \\ \frac{u'(c_t)}{u'(c_{t+1})} = \beta[f'(k_{t+1}) + (1-\delta)] \Rightarrow 1 = \rho[F'(k^*) + (1-\delta)] \Rightarrow F'(k^*) = \rho + \delta \Rightarrow k_G^* > k^* \end{cases} \tag{230}$$

---

## 5.6 Dynamic inefficiency

Because of the micro-foundations (patience denoted as $\beta = \frac{1}{1+\rho}$) in Ramsey, there will be no over-accumulation of capital, avoiding dynamic inefficiency. Whereas in Solow model the inefficiency is inevitable. Somehow in OLG model, the dynamic inefficiency returns.

**Solow**

The model has key equations

$$\begin{cases} Y = F(K, L) \\ I = S = sY \\ L_{t+1} = (1+n)L_t \\ k_{t+1} = \frac{(1-\delta)k_t + sf(k_t)}{(1+n)} \iff \dot{k} = sf(k) - (n+\delta)k \end{cases} \tag{231}$$

For a certain production $y = k^a$ at steady state

use $k_{t+1} = \frac{(1-\delta)k_t + sf(k_t)}{1+n}$

$$\Rightarrow (1+n)k^* = (1-\delta)k^* + sf(k^*) \tag{232}$$

$$-(n+\delta)k^* = -sf(k^*) \tag{233}$$

$$f(k^*) - (n+\delta)k^* = (1-s)f(k^*) = c^* \tag{234}$$

$$\Rightarrow c^* = f(k^*) - (n+\delta)k^* \tag{235}$$

$$\text{To maximize welfare at steady state} \iff \max c* \iff FOC: k_g^* = (\frac{n+\delta}{\alpha})^{1-\alpha} \tag{236}$$

also at steady state

$$(1+n)k^* = (1-\delta)k^* + sf(k^*) \tag{237}$$

$$\Rightarrow k^* = (\frac{n+\delta}{s})^{1-\alpha} \tag{238}$$

for certain $\alpha$ and $s$ it could be

$$\Rightarrow k^* < k_g^* \tag{239}$$

meaning there is possible dynamic inefficiency in Solow model

**Ramsey**

Ramsey has no dynamic inefficiency, here's why

The model has three key equations

$$\begin{cases} L_t = (1+n)L_{t-1} \\ k_{t+1} - (1-\delta)k_t = f(k_t) - c_t \\ \frac{u'(c_t)}{u'(c_{t+1})} = \beta[f'(k_{t+1}) + (1-\delta)] \end{cases} \tag{240}$$

Put $k_{t+1} - (1-\delta)k_t = f(k_t) - c_t$ at steady state

$$c^* = f(k^*) - k^* + (1-\delta)k^* = f(k^*) - \delta k^* \tag{241}$$

$$\text{To maxmize welfare at steady state} \Rightarrow \text{FOC: } f'(k_G^*) = \delta \tag{242}$$

Put $\frac{u'(c_t)}{u'(c_{t+1})} = \beta[f'(k_{t+1}) + (1-\delta)]$ at steady state resulting

$$\Rightarrow 1 = \beta[f'(k^*) + (1-\delta)] = \frac{1}{1+\rho}[f'(k^*) + (1-\delta)] \tag{243}$$

$$\Rightarrow 1 + \rho = f'(k^*) + 1 - \delta \tag{244}$$

$$\Rightarrow f'(k^*) = \rho + \delta < f'(k_g^*) \tag{245}$$

$$\Rightarrow k_G^* > k^* \tag{246}$$

Capital per capita at steady state is lower than $k_{gold}$, meaning it is dynamic efficient.

**OLG**

OLG in central planner form:

$$\begin{aligned} &\max \sum_{t=0}^{\infty} (\beta_s)^t L_t[u(c_{1t}) + \beta u(c_{2,t+1})] \text{ ($\beta_s$ is the discount factor)} \\ &s.t.\ c_{1t} + \frac{c_{2t}}{1+n} + (1+n)k_{t+1} = f(k_t) \end{aligned} \tag{247}$$

Equation of motion of capital per capita at the steady states

$$c^* \equiv c_1^* + \frac{c_2^*}{1+n} = f(k^*) - (1+n)k^* \tag{248}$$

Since we are maximizing welfare at steady state, let $\frac{\partial c^*}{\partial k^*} = 0$ to acquire first order condition

$$\Rightarrow f'(k_G^*) = 1 + n \tag{249}$$

using production function $y = k^\alpha$ at equilibrium

$$f'(k^*) = \alpha(k^*)^{\alpha-1} \tag{250}$$

$$= \alpha\{[\frac{\beta(1-\alpha)}{(1+\beta)(1+n)}]^{\frac{1}{1-\alpha}}\}^{\alpha-1} \tag{251}$$

$$= \frac{\alpha}{1-\alpha}\frac{1+\beta}{\beta}(1+n) \tag{252}$$

for certain value of $\alpha$

$$f'(k^*) < 1 + n \tag{253}$$
$$\Rightarrow f'(k^*) < f'(k_G^*) \tag{254}$$
$$\Rightarrow k^* > k_G^* \tag{255}$$

Meaning there can be over-accumulation of capital $\iff$ suffices dynamic inefficiency possible in OLG model.

**Conclusion**

- Due to ad hoc given $s$, Solow model is unable to assure dynamic efficiency.
- Due to endogenous saving decision and discount factor $\beta$, Ramsey model is able to achieve dynamic efficiency.
- Due to limited life span of individuals, OLG model fails to achieve dynamic efficiency.

## 5.7 Population, pensions, and public debt

*Planned section.* Extend the two-generation benchmark to population change, pension arrangements, and government debt. State the conditions for dynamic inefficiency before comparing intergenerational transfers.

# 6 Endogenous Growth and Resource Allocation

*Planned chapter.*

## 6.1 The AK benchmark and human capital

Ask when reproducible factors can sustain growth without exogenous technological progress. Compare the roles of returns to capital and human-capital accumulation.

## 6.2 Ideas, innovation, and incentives

Introduce nonrival ideas and incentives to innovate. Separate the private return to innovation from its social return and discuss the policy implications.

## 6.3 Allocation and aggregate productivity

Explain how differences in productivity and financing conditions affect the allocation of capital. Connect allocation to aggregate output and growth.

# Part II
# Fluctuations

How do shocks change production, spending, employment, and prices? We begin with a real benchmark, introduce nominal rigidities, and then examine labor-market adjustment. Part IV adds financial and distributional mechanisms and develops the policy comparison.

# 7 Real Business Cycle Model

Originated from Kyland and Prescott (1982), Long and Plosser (1983), RBC model shows great empirical importance. A standard RBC model[3] is in nature a Ramsey model with

- Endogenous (but constant in the long run, i.e. $n = 0$) labor supply
- Stochastic productivity (i.e. real shocks)

**Note 11** (Two principles of the RBC theory of business cycles)**.** *1. Money is of little importance in business cycles.*
*2. Business cycles are created by rational agents responding optimally to real (not nominal) shocks - mostly fluctuations in productivity growth, but also fluctuations in government purchases, import prices, or preferences.*

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## 7.1 UMP of household

A typical household entails product-consuming for $C_t$ and labor-providing for $N_t$. Only in this model, the agent value leisure $L_t = 1 - N_t$.
The UMP in RBC model is

$$\max_{C_t, N_t, B_t} E_t \sum_{t=0}^{\infty} \beta^t u(C_t, N_t) \tag{256}$$

$$s.t.\ P_t C_t + Q_t B_t \leqslant B_{t-1} + W_t N_t + D_t, \text{ where } \begin{cases} \beta \in (0,1) \\ Q_t = \frac{1}{1+i_t} \text{ is the price per bond bought today} \end{cases} \tag{257}$$

Form a Lagrangian and acquire FOCs

$$\mathscr{L} = E_0 \sum_{t=0}^{\infty} \beta^t [u(C_t, N_t) + \lambda_t (P_t C_t + Q_t B_t - B_{t-1} - W_t N_t - D_t)] \tag{258}$$

$$\Rightarrow \begin{cases} \frac{\partial \mathscr{L}}{\partial C_t} = 0 \Rightarrow \beta^t [u_{C_t} - \lambda_t P_t] = 0 \Rightarrow \frac{u_{C_t}}{E_t U_{c,t+1}} = \frac{\lambda_t}{E_t \lambda_{t+1}} \cdot \frac{P_t}{E_t P_{t+1}} \\ \frac{\partial \mathscr{L}}{\partial N_t} = 0 \Rightarrow \beta^t [u_{N_t} + \lambda_t W_t] = 0 \\ \frac{\partial \mathscr{L}}{\partial B_t} = 0 \Rightarrow -\lambda_t Q_t \beta^t + E_t \lambda_{t+1} \beta^{t+1} = 0 \Rightarrow \beta \frac{E_t \lambda_{t+1}}{\lambda_t} = Q_t \Rightarrow \frac{1}{Q_t} = 1 + i_t \end{cases} \tag{259}$$

using these FOCs

$$\begin{cases} \beta^t [u_{C_t} - \lambda_t P_t] = 0 \\ \beta^t [u_{N_t} + \lambda_t W_t] = 0 \end{cases} \Rightarrow \text{Labor supply equation} - \frac{u_{N_t}}{u_{C_t}} = \frac{W_t}{P_t} \tag{260}$$

$$\text{Also} \begin{cases} \frac{u_{C_t}}{E_t U_{c,t+1}} = \frac{\lambda_t}{E_t \lambda_{t+1}} \cdot \frac{P_t}{E_t P_{t+1}} \\ \beta \frac{E_t \lambda_{t+1}}{\lambda_t} = Q_t \end{cases} \Rightarrow \text{Euler's equation } \frac{Q_t}{\beta} = E_t \cdot \frac{\frac{U_{c,t+1}}{P_{t+1}}}{\frac{U_{c,t}}{P_t}} \tag{261}$$

[3]For a sound review over the topic of RBC, see Rebelo Sergio. 2005. "Real Business Cycle Models: Past, Present, and Future." The Scandinavian Journal of Economics.

e.g. Let utility function be specific

$$u(C_t, N_t) = \begin{cases} \frac{c_t^{1-\sigma}-1}{1-\sigma} - \frac{N_t^{1+\varphi}}{1+\varphi}, \sigma \neq 1 \\ \log C_t - \frac{N_t^{1+\varphi}}{1+\varphi}, \sigma = 1 \end{cases} \tag{262}$$

$$\Rightarrow \begin{cases} \frac{W_t}{P_t} = C_t^{\sigma} N_t^{\varphi} \\ Q_t = \beta E_t \cdot (\frac{C_{t+1}}{C_t})^{-\sigma} \cdot \frac{P_t}{P_{t+1}} \end{cases} \tag{263}$$

---

## 7.2 PMP of firm

$$\min_{N_t} W_t N_t \Leftrightarrow \max_{Y_t} P_t Y_t - W_t N_t \tag{264}$$

$$s.t. Y_t = A_t K_t^{\alpha} N_t^{1-\alpha}, \alpha \in [0, 1) \tag{265}$$

$$\text{where } \begin{cases} Y_t = A_t W_t^{1-\alpha} \\ A_t = A_{t-1}^{\rho\alpha} \epsilon_t^{\alpha}, \epsilon_t^{\alpha} \overset{iid}{\sim} N(0, \sigma_{\alpha}^{\epsilon}) \text{ (technology shocks)} \end{cases} \tag{266}$$

$$\Rightarrow \text{F.O.C.: } \frac{W_t}{P_t} = (1-\alpha) A_t N_t^{-\alpha} \text{ (labor demand equation)} \tag{267}$$

---

## 7.3 Economy described by RBC

The whole economy system is described as the balance of demand and supply

$$\text{AS} \begin{cases} Y_t = A_t N_t^{1-\alpha} \text{ (total production function)}, A_t = A_{t-1}^{\rho a} e^{\epsilon_t^a} \text{ (technology shocks)} \\ \frac{W_t}{P_t} = C_t^{\sigma} N_t^{\varphi} \text{ (labor supply from UMP)} \\ \frac{w_t}{P_t} = (1-\alpha) A_t N_t^{-\alpha} \text{ (labor demand from PMP)} \end{cases} \tag{268}$$

$$\text{AD} \begin{cases} Y_t = C_t + I_t \text{ (accouting equation)} \\ Q_t = \beta E_t (\frac{C_{t+1}}{C_t})^{-a} \frac{P_t}{P_{t+1}} \text{ (Euler equation from UMP)} \\ \frac{M_t}{P_t} = \frac{Y_t}{Q_t^{-n}} \text{ (demand of liquidity)} \end{cases} \tag{269}$$

---

## 7.4 Log-linearization

Two methods we can transform. 1) The first is

$$\hat{x}_t = \ln x_t - \ln x^* \tag{270}$$
$$= \ln|\frac{x_t}{x^*}| \tag{271}$$
$$= \ln|1 + \frac{x_t - x^*}{x^*}| \text{ (percentage of deviation from stable state)} \tag{272}$$
$$\approx \ln|1 + \frac{1}{x^*}(x_t - x^*)| \tag{273}$$
$$= \frac{x_t - x^*}{x^*} \tag{274}$$

2) Another approach would be

$$\ln x^* = lnx_t - \tilde{x_t} \Rightarrow \ln x_t = lnx^* + \tilde{x_t} \tag{275}$$
$$\Rightarrow e^{\ln x_t} = x_t = e^{nx^* + \tilde{x_t}} = x^* e^{\hat{x_t}} \approx x^*(1 + \hat{x}_t) \tag{276}$$

Now we turn to the linearization of RBC model

$$\begin{cases} y_t = a_t + (1-\alpha)n_t \\ y = a + (1-\alpha)n \end{cases} \Rightarrow \hat{y}_t = \hat{a}_t + (1-\alpha)\hat{n}_t \tag{277}$$

$$\Rightarrow \begin{cases} \hat{w}_t - \hat{p}_t = \hat{a}_t + (1-\alpha)\hat{n}_t \\ \hat{w}_t - \hat{p}_t = a\hat{c}_t + \rho\hat{n}_t \end{cases} \Rightarrow \hat{y}_t = \hat{c}_t,\ m_t - \hat{p}_t = \hat{y}_t - \hat{n}_{it} \tag{278}$$

$$Y_t = C_t + I_t \tag{279}$$
$$Y \cdot e^{\hat{y}_t} = C \cdot e^{\hat{c}_t} + I \cdot e^{\hat{i}_t} \tag{280}$$
$$Y(1 + \hat{y}_t) = C(1 + \hat{c}_t) + I(1 + \hat{i}_t) \tag{281}$$
$$\hat{y}_t = \frac{C}{Y}\hat{c}_t + \frac{I}{Y}\hat{i}_t \tag{282}$$
$$a_t = \rho_t a_{t-1} + \epsilon_t^a \tag{283}$$
$$1 = E_t e^{\ln\beta + + \ln\frac{1}{Q_t} - a(c_{t+1} - c_t) - (P_{t+1} - P_t)} \tag{284}$$
$$1 = E_t e^{\ln\beta + + \ln\frac{1}{Q_t} - a\triangle c_{t+1} - \pi_{t+1}} \tag{285}$$
$$1 = E_t I[1 + (i_t - i) - a(\triangle c_{t+1} - \triangle c_t] - (\pi_{t+1} - \pi_t)] \tag{286}$$

Therefore a RBC model can be rewritten in linearized log equations

$$\text{AS:} \begin{cases} \hat{y}_t = a_t + (1-\alpha)\hat{n}_t \text{ (output)} \\ \hat{w}_t - \hat{p}_t = a_t - \alpha\hat{n}_t \text{ (labor demand)} \\ \hat{w}_t - \hat{p}_t = ac_t + \varphi\hat{n}_t \text{ (labor supply)} \\ \hat{a}_t = \rho_a \hat{a}_{t-1} + \epsilon_t^a \text{ (technology)} \end{cases} \tag{287}$$

$$\text{AD:} \begin{cases} \hat{y}_t = \hat{c}_t \text{ (product market clearing)} \\ \hat{c}_t = E_t\hat{c}_{t+1} - \frac{1}{a}(\hat{i}_t - E_t\hat{\pi}_{t+1}) \\ \hat{m}_t - \hat{p}_t = \hat{y}_t - \eta \cdot \hat{i}_t \text{ (money demand equation; } \eta \text{ is interest rate elasticity)} \\ \hat{r}_t = i_t - E_t\hat{\pi}_{t+1} \text{ (Fisherian equation)} \end{cases} \tag{288}$$

**Note 12** (Exceptation for Expectation)**.** *Notice this method does not apply to equations involving expectations.*

**Note 13** (RBC has neutrality of money)**.** *The final answer of this standard RBC model is not relevant to any price variable, inferring the model being money-neutral.*

## 7.5 Capital accumulation and investment

*Planned section.* Extend the current fixed-capital benchmark to endogenous capital accumulation. Derive investment dynamics and distinguish the simplified benchmark from a full RBC model with a changing capital stock.

# 8 New Keynesian Model

DNK is short for Dynamic New Keynesian[4]
This New keynesian Model contains elements being

- dispersive economy, a continuum of agents
- perfect information
- rational expectations, i.e. $E_t(X_{t+1}) = E(X_{t+1}|I_t)$
- perfect competitive labor market
- monopolisitc competitive product market, heterogenous products, nominal rigidities

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## 8.1 UMP of household

**Two stages of optimization**
Goal of UMP is to acquire DIS, which represents demand. Household face two stages of optimization. Since goods are heterogenous, UMP begins by choosing items $i \in I$.
Stage I: choosing goods
One way to choose products is by method of maximization.

$$\begin{aligned} &\max_{C_{it}}(\int_0^1 C_{it}^{\frac{\epsilon-1}{\epsilon}} di)^{\frac{\epsilon}{\epsilon-1}} \\ &s.t. \int_0^1 P_{it}C_{it}di \leqslant Z_t \end{aligned} \tag{289}$$

, in which the $\epsilon$ stands for current substitute elasticity and $\frac{\epsilon-1}{\epsilon}$ is contemporaneous elasticity of substitution. And $Z_t$ stands for endowment.
The other way to choose products is by method of minimization. In this case Lagrangian multiplier could be shadow price.

$$\begin{aligned} &\min \int_0^1 P_{it}C_{it}di \\ &s.t.(\int_0^1 C_{it}^{\frac{\epsilon-1}{\epsilon}} di)^{\frac{\epsilon}{\epsilon-1}} \geqslant C_t \end{aligned} \tag{290}$$

Form a Lagrangian, we would have

$$\underset{\{C_{it}\}}{\mathscr{L}} = \int_0^1 P_{it}C_{it}di + P_t[(\int_0^1 C_{it}^{\frac{\epsilon-1}{\epsilon}} di)^{\frac{\epsilon}{\epsilon-1}} - C_t] \tag{291}$$

$$\Rightarrow P_{it} = P_t \frac{\epsilon}{\epsilon-1}((\int_0^1 C_{it}^{\frac{\epsilon-1}{\epsilon}} di)^{\frac{\epsilon}{\epsilon-1}-1}) \frac{\epsilon-1}{\epsilon} C_{it}^{\frac{\epsilon-1}{\epsilon}-1} \tag{292}$$

$$\Rightarrow P_{it} = P_t[(\int_0^1 C_{it}^{\frac{\epsilon-1}{\epsilon}} di)^{\frac{\epsilon}{\epsilon-1}})^{\frac{1}{\epsilon}}]C_{it}^{\frac{-1}{\epsilon}} \tag{293}$$

$$\Rightarrow \frac{P_{it}}{P_t} = (\frac{C_{it}}{C_t})^{\frac{-1}{\epsilon}} \tag{294}$$

$$\Rightarrow C_{it} = (\frac{P_{it}}{P_t})^{-\epsilon} C_t \text{ (the demand curve)} \tag{295}$$

[4]This chapter is mainly based on the Gali, 2015.

From $\int_0^1 P_{it}C_{it}di = P_tC_t$ we have

$$\Rightarrow \int_0^1 P_{it}(\frac{P_{it}}{P_t})^{-\epsilon}C_t di = P_tC_t \tag{296}$$

$$\Rightarrow \int_0^1 P_{it}(\frac{P_{it}}{P_t})^{-\epsilon} di = P_t \tag{297}$$

$$\Rightarrow P_t = (\int_0^1 P_{it}^{1-\epsilon} di)^{\frac{1}{1-\epsilon}} \text{ (the expression of total price level)} \tag{298}$$

Apparently $\begin{cases} \max\limits_{C_{it}} (\frac{1}{I}\sum\limits_{i=1}^{I} C_{it}^{\frac{1}{1+n_t}})^{1+n_t} = C_t \\ s.t. \sum\limits_{i=1}^{I} P_{it}C_{it} \leqslant Z_t \end{cases}$ and $\begin{cases} \min\limits_{C_{it}} \sum\limits_{i=1}^{I} P_{it}C_{it} \\ s.t. (\frac{1}{I}\sum\limits_{i=1}^{I} C_{it}^{\frac{1}{1+n_t}})^{1+n_t}) \geqslant C_t \end{cases}$ are the same. But as mentioned before, the minimization approach is commonly prefered.
Using either approach, we can conduct the result that demand curve and expression of total price level being

$$C_{it} = (I \cdot \frac{P_{it}}{P_t})^{\frac{1+n_t}{-\wedge_t}} C_t \tag{299}$$

$$P_t = I(\frac{1}{I}\sum_{i=1}^{I} P_{it}^{\frac{1}{-\wedge_t}})^{-\wedge_t} \tag{300}$$

After picking products, we enter stage II, UMP at period $t$

$$\begin{aligned} &\max_{C_t, N_t, \frac{M_t}{P_t}} E_0 \sum_{t=0}^{\infty} \beta^t u\left(C_t, N_t, \frac{M_t}{P_t}\right) \leftarrow \text{(MIU)} \\ &s.t. \int_0^1 P_{it}C_{it}di + M_t + B_t \leqslant M_{t-1} + (1+i_{t-1})B_{t-1} + \int_o^1 W_{it}N_{it}di + T_t \end{aligned} \tag{301}$$

constraint of this UMP can be rewritten as

$$\begin{aligned} &\max_{C_t, N_t, \frac{M_t}{P_t}} E_0 \sum_{t=0}^{\infty} \beta^t u\left(C_t, N_t, \frac{M_t}{P_t}\right) \\ &s.t.\ P_tC_t + M_t + B_t \leqslant M_{t-1} + (1+i_{t-1})B_{t-1} + W_tN_t + T_t \end{aligned} \tag{302}$$

**Note 14** (Subscript $it$ and $t$)**.** $C_{it} \neq C_t \Leftrightarrow$ *Monopolistic competitive goods market;* $N_{it} = N_t \Leftrightarrow$ *Perfect competitve labor market*

Form the Lagrangian

$$\mathscr{L} = E_0 \sum_{t=0}^{\infty} \beta^t \{u(C_t, N_t, \frac{M_t}{P_t}) + \lambda_t[P_tC_t + M_t + B_t - M_{t-1} - (1+i_{t-1})B_{t-1} - W_tN_t + T_t]\} \tag{303}$$

Solving this will deliver the F.O.C.s., resulting

$$\Rightarrow \begin{cases} Q_t = \beta E_t \cdot (\frac{u_{C,t+1}}{u_{C,t}} \frac{P_t}{P_{t+1}}) \\ \frac{W_t}{P_t} = -\frac{u_{N_t}}{U_{C_t}} \\ 1 - Q_t = u_{\frac{M_t}{P_t}} \end{cases} \tag{304}$$

This result is basically same as in RBC model with one more equation since money is added to the utility function.

**Linearization**

Linearization specificly focusing on the third equation of the second equation, as the other two are identical in RBC model's linearization.

$$1 - \frac{1}{1+i_t} = C_t^{\sigma}(\frac{M_t}{P_t})^{-\nu} \text{ (using } Q_t\text{'s defintion)} \tag{305}$$

$$\Rightarrow \frac{i_t}{1+i_t} = C_t^{\sigma}(\frac{M_t}{P_t})^{-\nu} \tag{306}$$

Taking logs

$$\log \frac{i_t}{1+i_t} = \sigma c_t - \nu(m_t - p_t) \text{ (hereafter a lowercase letter } x \text{ denotes } \log x) \tag{307}$$

$$\log \frac{i}{1+i} = \sigma c - \nu(m - p) \tag{308}$$

$$\Rightarrow \log(\frac{i_t}{1+i_t}) - \log(\frac{i}{1+i}) = \sigma \hat{c}_t - \nu(m_t - \hat{p}_t) \tag{309}$$

$$\Rightarrow \nu(m_t - \hat{p}_t) = \sigma \hat{c}_t + \log(\frac{1+i_t}{i_t}) - \log(\frac{1+i}{i}) \tag{310}$$

$$\Rightarrow \nu(m_t - \hat{p}_t) = \sigma \hat{c}_t + [\log \frac{(\frac{1+i}{i}) + 1}{\frac{1+i}{i}} \frac{i - (1+i)}{i^2}(i_t - i) - \log(\frac{1+i}{i})] \tag{311}$$

$$\Rightarrow \nu(m_t - \hat{p}_t) = \sigma \hat{c}_t + \frac{i}{i+1}\frac{i-(1+i)}{i}\frac{i_t - i}{i} \tag{312}$$

$$\Rightarrow \nu(m_t - \hat{p}_t) = \sigma \hat{c}_t - \frac{1}{1+i}\hat{i}_t \tag{313}$$

$$\Rightarrow m_t - \hat{p}_t = \frac{\sigma}{\nu}\hat{c}_t - \frac{1}{(1+i)\nu}\hat{i}_t \tag{314}$$

$$\Rightarrow m_t - \hat{p}_t = \hat{c}_t - \eta\hat{i}_t \tag{315}$$

**Note 15** (Must-be in exam: monopolistic competition vs perfect competition)**.** *Zero market power to finite market power, meaning infinite elasticiy to non-zero positive elasticity.*

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## 8.2 PMP of firm

**Note 16** (Two features DNK firm has)**.** *Monopolistic competitive goods market and stikcy price.*

### 8.2.1 Price-setting behavioral equation

We first use PMP to acquire price-setting equation then linearize it. Next we use price-setting equation to derive NKPC.

**PMP and price-setting behavioral equation**

Set base period on the intial period $t = 0$ for simplicity

$$\begin{aligned}&\max_{P_{i0}} \pi_0^n = E_0 \sum_{t=0}^{\infty} Q_{0,t} \cdot (1-\theta)^t \{P_{i0} Y_{it|0} - [TC_{it|0}^n(Y_{it|0})]\} \\ &s.t.\ Y_{it|0} = (\frac{P_{i0}}{P_t})^{-\epsilon} Y_t \text{ (since } C_t = Y_t \text{ at equilbrium, the constraint is essentially demand curve)}\end{aligned} \tag{316}$$

$|0$ means evaluated at prices from period $0$
$Y_{it}$ is total sales of firm $i$ at period $t$
$TC$ is a function of nominal total cost
$Q_{0,t}$ means discount rate from period $0$ to period $t$
$\theta$ means the probability of price adjusting, therefore $1 - \theta$ means the probablity to stay put. It's sticky price. [5]

Solving the PMP by putting demand curve into profit function, it would be

$$\pi_0^n = E_0 \sum_{t=0}^{\infty} Q_{0,t} \cdot (1-\theta)^t \{P_{i0} (\frac{P_{i0}}{P_t})^{-\epsilon} Y_t - [TC_{it|0}^n((\frac{P_{i0}}{P_t})^{-\epsilon} Y_t)]\} \tag{317}$$

then let the F.O.C. be $0$ (fig 1,2)

$$\frac{\partial \pi_0^n}{\partial P_{i0}} = 0 \tag{318}$$

$$\Rightarrow E_0 \sum_{t=0}^{\infty} Q_{0,t} \cdot (1-\theta)^t \{(1-\theta)(\frac{P_{i0}}{P_t})^{-\epsilon} Y_t - [\frac{\partial TC_{it|0}^n}{\partial P_{it|0}} \frac{\partial P_{it|0}}{\partial P_{i0}}] = 0 \tag{319}$$

$$\Rightarrow E_0 \sum_{t=0}^{\infty} (1-\theta)^t Q_{0,t} [(1-\epsilon)(\frac{P_{i0}}{P_t})^{-\epsilon} Y_t - MC_{it|0}^n(-\epsilon)(\frac{P_{i0}}{P_t})^{-\epsilon-1} Y_t \frac{1}{P_t}] = 0 \tag{320}$$

$$\Rightarrow E_0 \sum_{t=0}^{\infty} (1-\theta)^t Q_{0,t} [(1-\epsilon) Y_t - MC_{it|0}^n(-\epsilon)(\frac{P_{i0}}{P_t})^{-\epsilon-1} Y_t \frac{1}{P_t}] = 0 \tag{321}$$

$$\Rightarrow E_0 \sum_{t=0}^{\infty} (1-\theta)^t Q_{0,t} [(1-\epsilon) Y_{it|0} + \frac{\epsilon}{P_t} MC_{it|0}^n (\frac{P_{i0}}{P_t})^{-1}] = 0 \tag{322}$$

$$\Rightarrow E_0 \sum_{t=0}^{\infty} (1-\theta)^t Q_{0,t} Y_{it|0} [(1-\epsilon) + \epsilon MC_{it|0}^n \cdot (P_{i0})^{-1}] = 0 \tag{323}$$

$$\Rightarrow E_0 \sum_{t=0}^{\infty} (1-\theta)^t \cdot Q_{0,t} \cdot Y_{it|0} \cdot (1-\epsilon) = -E_0 \sum_{t=0}^{\infty} (1-\theta)^t \cdot Q_{0,t} \cdot Y_{it|0} \cdot \epsilon \cdot MC_{it|0}^n \cdot (P_{i0})^{-1} \tag{324}$$

$$\Rightarrow (1-\epsilon) E_0 \sum_{t=0}^{\infty} (1-\theta)^t \cdot Q_{0,t} \cdot Y_{it|0} = -\epsilon (P_{i0})^{-1} E_0 \sum_{t=0}^{\infty} (1-\theta)^t \cdot Q_{0,t} \cdot Y_{it|0} \cdot MC_{it|0}^n \tag{325}$$

restore $Q_{0t}$ and $Y_{it|0}$ back to demand curve

$$\Rightarrow (\epsilon - 1) E_0 \sum_{t=0}^{\infty} (1-\theta)^t \beta^t [(\frac{Y_0}{Y_t})^{\rho} \frac{P_0}{P_t}] (\frac{P_{i0}}{P_t})^{-\epsilon} Y_t = (P_{i0})^{-1} E_0 \sum_{t=0}^{\infty} (1-\theta)^t \beta^t [(\frac{Y_0}{Y_t})^{\rho} \frac{P_0}{P_t}] (\frac{P_{i0}}{P_t})^{-\epsilon} Y_t \cdot MC_{it|0}^n \tag{326}$$

[5] For more of price adjusting, see in Calvo, 1983 for uncertain version; Taylor, 1980 for certain version.

take $P_{i0}$ and $P_0$ out since it can be treated as constants

$$\Rightarrow (\epsilon-1)\underline{P_{i0}^{-\epsilon}P_0Y_o^{\sigma}}E_0\sum_{t=0}^{\infty}(1-\theta)^t\beta^tP_t^{\epsilon-1}Y_t^{1-\rho}=\epsilon\underline{P_{i0}^{-1}P_{i0}^{-\epsilon}P_0Y_0^{\rho}}E_0\sum_{t=0}^{\infty}(1-\theta)^t\beta^tP_t^{\epsilon-1}Y_t^{1-\sigma}\cdot MC_{it|0}^n \tag{327}$$

eliminates the underlined common factors on RH and LH

$$\Rightarrow (\epsilon-1)E_0\sum_{t=0}^{\infty}(1-\theta)^t\beta^tP_t^{\epsilon-1}Y_t^{1-\rho}=\epsilon E_0\sum_{t=0}^{\infty}(1-\theta)^t\beta^tP_t^{\epsilon-1}Y_t^{1-\sigma}\cdot MC_{it|0}^n \tag{328}$$

$$\Rightarrow P_{i0}^*=\frac{\epsilon}{\epsilon-1}\frac{E_0\sum\limits_{t=0}^{\infty}(1-\theta)^t\beta^tP_t^{\epsilon-1}Y_t^{1-\sigma}\cdot MC_{it|0}^n}{E_0\sum\limits_{t=0}^{\infty}(1-\theta)^t\beta^tP_t^{\epsilon-1}Y_t^{1-\sigma}}\text{ (sticky price form price-setting equation)} \tag{329}$$

Also acquiring optimal price $p^*$, and $\frac{\epsilon}{\epsilon-1}$ is cost mutltiplier

$$\Rightarrow P_{i0}^*=P_0^*=\frac{\epsilon}{\epsilon-1}\frac{[(1-\epsilon)\beta]^0P_o^{\epsilon-1}Y_0^{1-\sigma}MC_{0|0}^n}{(1-\theta)^0\beta^0P_0^{\epsilon-1}Y_0^{1-\sigma}}=\frac{\epsilon}{\epsilon-1}MC_{0|0}^n\text{ (elastic form price-setting equation)} \tag{330}$$

For elastic form, the intuition is that firms are adjusting price at any time so there is no need to observe profit based on any specific period. So no summation symbol.

**Question 4** (what if price is elastic?)**.**

elasticity means adjusting price at all time. it means price sticky price is $0 \Leftrightarrow 1-\theta=0$.

**Note 17.** *monopolistic competition and elasticity are not the same thing*

next try the period-$t$-based version in textbook (pic 3)

$$P_t^*=\frac{\epsilon}{\epsilon-1}\frac{E_t\sum\limits_{k=0}^{\infty}(1-\theta)^k\beta^{t+k}P_{t+k}^{\epsilon-1}Y_{t+k}^{1-\sigma}\cdot MC_{t+k|t}^n}{E_t\sum\limits_{k=0}^{\infty}(1-\theta)^k\beta^tP_{t+k}^{\epsilon-1}Y_{t+k}^{1-\sigma}} \tag{331}$$

(here the subscript $i$ is gone because all firms are making unanimous decisions)

$$P_t^*=\frac{\epsilon}{\epsilon-1}MC_{t|t}^n \tag{332}$$

**Note 18** (CRTS in DNK)**.** *if production function is CRTS, then all firms would also be making unanimous decisions regardless of time period position*

**log-linearization of sticky price form price-setting behavioral equation**
the method is still first order Taylor expansion, though bit trickier (pic 3, 4, 5)

dividing $P_{t-1}$ on both side, and try seperately

$$\underbrace{\frac{P_t^*}{P_{t-1}} E_t \sum_{k=0}^{\infty} (1-\theta)^k \beta^t P_{t+k}^{\epsilon-1} Y_{t+k}^{1-\sigma}}_{\text{LHS}} = \underbrace{\frac{1}{P_{t-1}} \frac{\epsilon}{\epsilon-1} E_t \sum_{k=0}^{\infty} (1-\theta)^k \beta^{t+k} P_{t+k}^{\epsilon-1} Y_{t+k}^{1-\sigma} \cdot MC_{t+k|t}^n}_{\text{RHS}} \tag{333}$$

LHS mutil-variate first order Taylor expansion, the constant term is LHS at steady state where all $P$ are equal

$$\begin{aligned} \text{LHS's 1st order Taylor expansion} = & \sum_{k=0}^{\infty} [(1-\theta)\beta]^k P^{\epsilon-1} Y^{1-\sigma} \\ & + \frac{1}{p} E_t \sum_{k=0}^{\infty} [(1-\theta)\beta]^k P^{\epsilon} Y^{1-\sigma} (\frac{P_t^* - P}{P}) \\ & - \frac{P}{P^2} E_t \sum_{k=0}^{\infty} [(1-\theta)\beta]^k P^{\epsilon-1} Y^{1-\sigma} (\frac{P_{t-1} - P}{P}) \\ & + (\epsilon-1) E_t \sum_{k=0}^{\infty} [(1-\theta)\beta]^k P^{\epsilon-1} Y^{1-\sigma} (\frac{P_{t+k} - P}{P}) \\ & + (1-\sigma) E_t \sum_{k=0}^{\infty} [(1-\theta)\beta]^k P^{\epsilon-1} Y^{1-\sigma} (\frac{Y_{t+k} - Y}{Y}) \end{aligned} \tag{334}$$

Like before, we use lowercase letters denotes log version

$$\Rightarrow \text{LHS} = P^{\epsilon-1} Y^{1-\sigma} E_t \sum_{k=0}^{\infty} [(1-\theta)\beta]^k P^{\epsilon-1} Y^{1-\sigma} [1 + \hat{p}_t^* - \hat{p}_{t-1} + (\epsilon-1)\hat{p}_{t+k} + (1-\sigma)\hat{y}_{t+k}] \tag{335}$$

For RHS,

$$\text{RHS} = \cdots = \frac{\epsilon}{\epsilon-1} \frac{1}{P} P^{\epsilon-1} Y^{1-\sigma} E_t \sum_{k=0}^{\infty} [(1-\theta)\beta]^k P^{\epsilon-1} Y^{1-\sigma} [1 - \hat{p}_{t-1} + (\epsilon-1)\hat{p}_{t+k} + (1-\sigma)\hat{y}_{t+k} + \hat{mc}_{t+k|t}^n] \tag{336}$$

**Note 19** (First order Taylor expansion and steady state)**.** *First term (constant term) of 1st Taylor expansion is steady state*

**Question 5** (why do we do log-linearization)**.**

to simulate a curve with a line, so there would be a foundable or writable solution.

Now the two sides are log-linearized, let them be in the same equation

$$\underbrace{P^{\epsilon-1} Y^{1-\sigma} E_t \sum_{k=0}^{\infty} [(1-\theta)\beta]^k P^{\epsilon-1} Y^{1-\sigma} [1 + \hat{p}_t^* - \hat{p}_{t-1} + (\epsilon-1)\hat{p}_{t+k} + (1-\sigma)\hat{y}_{t+k}]}_{\text{log-linear LHS}} = \tag{337}$$

$$\underbrace{\frac{\epsilon}{\epsilon-1} \frac{1}{P} P^{\epsilon-1} Y^{1-\sigma} E_t \sum_{k=0}^{\infty} [(1-\theta)\beta]^k P^{\epsilon-1} Y^{1-\sigma} [1 - \hat{p}_{t-1} + (\epsilon-1)\hat{p}_{t+k} + (1-\sigma)\hat{y}_{t+k} + \hat{mc}_{t+k|t}^n]}_{\text{log-linear RHS}} \tag{338}$$

Use sticky form price-setting equation at steady state (which because of steadiness looks like elastic form), we have

$$\frac{MC^n}{P} = \frac{\epsilon - 1}{\epsilon} \tag{339}$$

Since $E_t$ is followed by an infinite series, it is meaningless. Eliminate it.

$$\Rightarrow \hat{p}_t^* \cancel{E_t} \sum_{k=0}^{\infty} [(1-\theta)\beta]^k = E_t \sum_{k=0}^{\infty} [(1-\theta)\beta]^k \hat{mc}_{t+k|t}^n \tag{340}$$

$$\Rightarrow \frac{\hat{p}_t^*}{1-(1-\theta)\beta} = E_t \sum_{k=0}^{\infty} [(1-\theta)\beta]^k MC_{t+k|t}^n \tag{341}$$

$$\Rightarrow \hat{p}_t^* = [1-(1-\theta)\beta] E_t \sum_{k=0}^{\infty} [(1-\theta)\beta]^k \hat{mc}_{t+k|t}^n \text{ (log-linear sticky price setting equation)} \tag{342}$$

Therefore we have the log-linear optimal-price-setting equation for monopolistic competition and sticky price features firms.

### 8.2.2 Philips curve

**First way to acquire Philips curve**
next we use this log-linearized price-setting equation to acquire NK Philips curve
originally PC is about inflation and employment
step I, inflation (p7,8,9)

$$P_t = I(\frac{1}{I}\sum_{i=1}^{I} P_{it}^{\frac{1}{-\wedge_t}})^{-\wedge_t} \text{ (expression of total price)} \tag{343}$$

$$= [\int_0^{\theta} (P_{it}^*)^{1-\epsilon} di + \int_{\theta}^{1} P_{t-1}^{1-\epsilon} di]^{\frac{1}{1-\epsilon}} \text{ ($\theta$ portion of firms don't adjust price)} \tag{344}$$

$$= [\theta (P_{it}^*)^{1-\epsilon} + (1-\theta) P_{t-1}^{1-\epsilon}]^{\frac{1}{1-\epsilon}} \text{ (expression of total price with $\theta$ firms adjusting price)} \tag{345}$$

$$\Rightarrow P_t^{1-\epsilon} = \theta (P_{it}^*)^{1-\epsilon} + (1-\theta) P_{t-1}^{1-\epsilon} \tag{346}$$

$$\Rightarrow (\frac{P_t}{P_{t-1}})^{1-\epsilon} = \theta (\frac{P_t^*}{P_{t-1}})^{1-\epsilon} + (1-\theta) \tag{347}$$

$$\Rightarrow \Pi^{1-\epsilon} = \theta (\frac{P_t^*}{P_{t-1}})^{1-\epsilon} + (1-\theta) \text{ (the capital letter } \Pi \text{ is for total inflation } \frac{P_t}{P_{t-1}}) \tag{348}$$

Then log-linearization by bi-variate first order Taylor expansion

$$\Rightarrow \Pi^{1-\epsilon} + (1-\epsilon)\Pi^{1-\epsilon}(\frac{\Pi_t - \Pi}{\Pi}) = \{\theta(\frac{P}{P})^{1-\epsilon} + \theta(1-\epsilon)(\frac{P}{P})^{1-\epsilon}[\frac{1}{P}(P_t^* - P) - \frac{P}{P^2}\frac{(P_{t-1} - P)}{P}]\}^{1-\theta} \tag{349}$$

Since first term of tay exp is steady state, making $\Pi^{1-\epsilon} = 1$

$$\Rightarrow 1 + (1-\epsilon)\hat{\Pi}_t = \theta + \theta(1-\epsilon)(\hat{P}_t^* - \hat{P}_{t-1}) + (1-\theta) \tag{350}$$

$$\Rightarrow \hat{\Pi}_t = \theta(\hat{P}_t^* - \hat{P}_{t-1}) \tag{351}$$

$$\Rightarrow \frac{1}{\theta}\hat{\Pi}_t = \hat{P}_t^* - \hat{P}_{t-1} \tag{352}$$

Using price-setting equation, which is related to marginal cost, then production. (making nominal price combination of real price and $\hat{p}_{t-1}$)

$$\Rightarrow \frac{1}{\theta}\hat{\Pi}_t = [1-(1-\theta)\beta]E_t \sum_{k=0}^{\infty}[(1-\theta)\beta]^k(\hat{mc}^r_{t+k|t} + \hat{p}_{t+k} - \hat{p}_{t-1}) \tag{353}$$

$$= \underbrace{[1-(1-\theta)\beta]E_t \sum_{k=0}^{\infty}[(1-\theta)\beta]^k \hat{mc}^r_{t+k|t}}_{A} + [1-(1-\theta)\beta]E_t \sum_{k=0}^{\infty}[(1-\theta)\beta]^k(\hat{p}_{t+k} - \hat{p}_{t-1}) \tag{354}$$

Let the untouched part be A for simplicity, open the sum so there will be expression of inflation in log-linearized form

$$= A + [1-(1-\theta)\beta]E_t \sum_{k=0}^{\infty}[(1-\theta)\beta]^k(\hat{p}_{t+k} - \hat{p}_{t-1}) \tag{355}$$

$$\begin{aligned} &= A + [1-(1-\theta)\beta]E_t\{(\hat{p}_t - \hat{p}_{t-1}) + [(1-\theta)\beta](\hat{p}_{t+1} - \hat{p}_t + \hat{p}_t - \hat{p}_{t-1}) \\ &\quad + [(1-\theta)\beta]^2(\hat{p}_{t+2} - \hat{p}_{t+1} + \hat{p}_{t+1} - \hat{p}_t + \hat{p}_t - \hat{p}_{t-1}) + \dots\} \end{aligned} \tag{356}$$

$$= A + [1-(1-\theta)\beta]E_t\{[(\hat{\pi}_t + [(1-\theta)\beta](\hat{\pi}_{t+1} + \hat{\pi}_t) + [(1-\theta)\beta]^2(\hat{\pi}_{t+2} + \hat{\pi}_{t+1} + \hat{\pi}_t) + \dots\} \tag{357}$$

$$\begin{aligned} &= A + E_t\{[(\hat{\pi}_t + [(1-\theta)\beta](\hat{\pi}_{t+1} + \hat{\pi}_t) + [(1-\theta)\beta]^2(\hat{\pi}_{t+2} + \hat{\pi}_{t+1} + \hat{\pi}_t) + \dots\} \\ &\quad - (1-\theta)\beta E_t\{[(\hat{\pi}_t + [(1-\theta)\beta](\hat{\pi}_{t+1} + \hat{\pi}_t) + [(1-\theta)\beta]^2(\hat{\pi}_{t+2} + \hat{\pi}_{t+1} + \hat{\pi}_t) + \dots\} \end{aligned}$$

(making elimination possible)

(358)

$$\begin{aligned} &= A + E_t\{\hat{\pi}_t + [(1-\theta)\beta](\hat{\pi}_{t+1} + \hat{\pi}_t) + [(1-\theta)\beta]^2(\hat{\pi}_{t+2}\hat{\pi}_{t+1} + \hat{\pi}_t) + \dots\} \\ &\quad - [(1-\theta)\beta]\{\hat{\pi}_t - [(1-\theta)\beta]^2 E_t(\hat{\pi}_{t+1} + \hat{\pi}_t) - \dots\} \end{aligned} \tag{359}$$

Make format simple by doing iteration form, the first term is when $k = 0$, but the second still starts from k=0 so there will be some parameters changed

$$= [1-(1-\theta)\beta]E_t \sum_{k=0}^{\infty} \hat{mc}^r_{t+k|t} + E_t \sum_{k=0}^{\infty} \hat{\pi}_{t+k+1} \tag{360}$$

$$= \underbrace{[1-(1-\theta)\beta]mc^r_{t+k|t} + \pi_t}_{k=0} + [1-(1-\theta)\beta]E_t \sum_{k=0}^{\infty}[(1-\theta)\beta]^{k+1}\hat{mc}^r_{t+k+1|t} + E_t \sum_{k=0}^{\infty}[(1-\theta)\beta]^{k+1}\hat{\pi}_{t+k} \tag{361}$$

$$= [1-[(1-\theta)\beta]]\hat{mc}^r_{t|t} + \hat{\pi}_t + [(1-\theta)\beta]\{\qquad\} \tag{362}$$

$$= B \text{ (back to recursive form and let it be B)} \tag{363}$$

$$\Rightarrow \frac{1}{\theta}\hat{\pi}_t = B \tag{364}$$

$$\Rightarrow \hat{\pi}_t = \underbrace{\frac{\theta[1-(1-\theta)\beta]}{1-\theta}\hat{mc}^r_t}_{\text{effect of current marginal cost}} + \underbrace{\beta E_t\hat{\pi}_{t+1}}_{\text{effect of expectation of future inflation}} \quad \text{(New Keynesian Philips Curve marginal cost version)} \tag{365}$$

**Question 6** (at which variable are we making derivative of Lagrangian function ?)**.**

Theoratically we can make derivative of any variable if possible but not proper.

NKPC has another version where marginal cost is substitued with real output, thus measuring the effect of output gap.

$$\begin{aligned} &\min \frac{W_t}{P_t} N_t \\ &s.t. Y_t = A_t N_t^{1-\alpha} \end{aligned} \tag{366}$$

Form a Lagrangian

$$\Rightarrow \mathscr{L} = \frac{W_t}{P_t} N_t + MC_t^r \cdot (Y_t - A_t N_t^{1-\alpha}) \text{ (shadow price can also be marginal cost)} \tag{367}$$

Take partial derivative of $N_t$ here since labor is decided by market, the firm only can control demand of labor

$$\Rightarrow \frac{\partial \mathscr{L}}{\partial N_t} = 0 \tag{368}$$

$$\Rightarrow \frac{W_t}{P_t} N_t = MC_t^r \cdot MPN_t \text{ , where } MPN_t = \frac{\partial Y_t}{\partial N_t} = (1-\alpha) A_t N_t^{-\alpha} \tag{369}$$

$$\Rightarrow MC_t^r = \frac{\frac{W_t}{P_t}}{MPN_t} \tag{370}$$

$$\hat{mc}_t^\sigma = \hat{w}_t - \hat{p}_t - (a_t - \alpha \hat{n}_t) \text{ (log-linearized, and using labor supply equation)} \tag{371}$$

$$= \sigma \hat{c}_t + \varphi \hat{n}_t - (a_t - \alpha \hat{n}_t) \tag{372}$$

$$= \sigma \hat{y}_t - a_t - (\varphi + \alpha) \frac{\hat{y}_t - a_t}{1-\alpha} \tag{373}$$

$$= \frac{\sigma(1-\alpha) + (\varphi+\alpha)}{1-\alpha} \hat{y}_t - \frac{1+\varphi}{1-\alpha} a_t \tag{374}$$

To acquire natural ratio of output, we use the fact that elastic marginal cost is zero.

$$\Rightarrow 0 = \hat{mc}_t^r = \frac{\sigma(1-\alpha) + (\varphi+\alpha)}{1-\alpha} \hat{y}_t^f - \frac{1+\varphi}{1-\alpha} a_t \text{ (upscript } f \text{ stands for flexible meaning elastic)} \tag{375}$$

$$\Rightarrow \frac{1+\varphi}{1-\alpha} a_t = \frac{\sigma(1-\alpha) + (\varphi+\alpha)}{1-\alpha} \hat{y}_t^f \tag{376}$$

$$\Rightarrow \hat{mc}_t^r = \frac{\sigma(1-\alpha) + (\varphi+\alpha)}{1-\alpha} ( \underbrace{\hat{y}_t - \hat{y}_t^f}_{\equiv \tilde{y}_t, \text{ output gap}} ) \text{ (put it back, having the gap of output } \hat{y}_t - y_t^f = \tilde{t}) \tag{377}$$

If we put the last equation back to NKPC marginal cost version, we connect natural output to inflation which is NKPC real output version.

New Keynesian Philips Curve marginal cost version:

$$\hat{\pi}_t = \underbrace{\frac{\theta[1-(1-\theta)\beta]}{1-\theta} \frac{\sigma(1-\alpha) + (\varphi+\alpha)}{1-\alpha} (\hat{y}_t - \hat{y}_t^f)}_{\text{effect of current output gap}} + \underbrace{\beta E_t \hat{\pi}_{t+1}}_{\text{effect of expectation of future inflation}} \tag{378}$$

another way to acquire the NKPC is to concern monopolistic competition, then sticky price.

$$\max_{P_{it}} \pi^r = \frac{P_{it}}{P_t} Y_{it} - \frac{W_t}{P_t} N_{it} \tag{379}$$

$$\Rightarrow \frac{P_{it}}{P_t} = \frac{\epsilon}{\epsilon - 1} MC_t^r \tag{380}$$

**Note 20** (indicator of monopolistic competition)**.**

$\frac{\epsilon}{\epsilon-1}$ means monopolistic competition
integral means perfect competitive labor supply for households
$\epsilon$ is subtitute elasticity

This is real price, where $\epsilon$ is subtitute elasticity. Here $MC_i$ comes without subscript $t$ is because of CRTS production function.

$$\Rightarrow \frac{P_{it}}{P_t} = \frac{\epsilon}{\epsilon - 1} \frac{W_t / P_t}{A_t} \tag{381}$$

$$= \frac{\epsilon}{\epsilon - 1} \frac{Y_t^\sigma N_t^\varphi}{A_t} \tag{382}$$

$$= \frac{\epsilon}{\epsilon - 1} \frac{Y_t^\sigma}{A_t} (\int_0^1 N_{it} di)^\varphi \text{ (because of goods market p.c.)} \tag{383}$$

$$= \frac{\epsilon}{\epsilon - 1} \frac{Y_t^\sigma}{A_t} [\int_0^1 (\frac{Y_{it}}{A_t}) di]^\varphi \text{ (based on CRTS production function)} \tag{384}$$

$$= \frac{\epsilon}{\epsilon - 1} \frac{Y_t^\sigma}{A_t^{1+\varphi}} [\int_0^1 (\frac{P_{it}}{P_t})^{-\epsilon} Y_t di]^\varphi \text{ (using demand curve and move } A_t \text{ out)} \tag{385}$$

$$= \frac{\epsilon}{\epsilon - 1} \frac{Y_t^\sigma}{A_t^{1+\varphi}} (\frac{P_{it}^*}{P_t})^{-\epsilon\varphi} \text{ (star means symmetric and at equil)} \tag{386}$$

Since at equilibrium all prices are same

$$P_{it}^* = P_t^* \Rightarrow \frac{P_{it}^*}{P_t} = \frac{\epsilon}{\epsilon - 1} \frac{Y_t^{\sigma+\varphi}}{A_t^{1+\varphi}} (\frac{P_t^*}{P_t})^{-\epsilon\varphi} \tag{387}$$

Let's consider a situation that

$$\text{If } P_t^* = P_t \text{ (meaning when price elastic)} \tag{388}$$

$$1 = \frac{\epsilon}{\epsilon - 1} \tag{389}$$

$$\Rightarrow Y_t^f \text{ or } \hat{y}_t^f \text{ (to be log-linearized)} \tag{390}$$

Taking logs

$$(1 + \epsilon\varphi)(P_t^* - P_t) = \ln \frac{\epsilon}{\epsilon - 1} - (1 + \varphi) a_t + (\sigma + \varphi) y_t \tag{391}$$

$$= (\sigma + \phi) y_t - (1 + \phi) a_t + \ln(\frac{\epsilon}{\epsilon - 1}) \text{ (change order)} \tag{392}$$

Change order for furthur preparation

$$\Rightarrow P_t^* = P_t + \frac{\sigma+\varphi}{1+\epsilon\varphi} y_t - \frac{1+\varphi}{1+\epsilon\varphi} a_t + \frac{1}{1+\epsilon\varphi} \ln(\frac{\epsilon}{\epsilon-1}) \tag{393}$$

$$\Rightarrow P^* = P + \frac{\sigma+\varphi}{1+\epsilon\varphi} y_t - 0 + \frac{1}{1+\epsilon\varphi} \ln(\frac{\epsilon}{\epsilon-1}) \text{ (steady state by definition)} \tag{394}$$

Two equations above deducted, having the log-linearized result; elastic price means $p_t^* = \hat{p}_t$

$$\hat{p}_t^* = \hat{p}_t + \frac{\sigma+\varphi}{1+\epsilon\varphi} \hat{y}_t - \frac{1+\varphi}{1+\epsilon\varphi} a_t \tag{395}$$

$$P_t^* = \hat{p}_t \text{ (elastic price gives } y \text{ a } f \text{ upscript)} \tag{396}$$

$$\Rightarrow 0 = \frac{\sigma+\varphi}{1+\epsilon\varphi} \hat{y}_t - \frac{1+\varphi}{1+\epsilon\varphi} a_t \tag{397}$$

$$\Rightarrow \hat{y}_t^f = \frac{1+\varphi}{\sigma+\varphi} a_t \tag{398}$$

here notice natural output is a function of technology

$$\Rightarrow \hat{p}_t^* = \hat{p}_t + \frac{\sigma+\varphi}{1+\epsilon\varphi} \hat{y}_t - \frac{\sigma+\varphi}{1+\epsilon\varphi} \hat{y}_t^f \tag{399}$$

$$\Rightarrow \hat{p}_t^* = \hat{p}_t + \gamma(\hat{y}_t - \hat{y}_t^f) \text{ ($\gamma$ denotes the bundle of coefficients)} \tag{400}$$

$$\Rightarrow \hat{p}_t^* = \hat{p}_t + \gamma\tilde{y}_t \text{ (acceptable price-setting euqation with only assumption of m.c.)} \tag{401}$$

6

**Note 21** (DNK model is perfect information rational expectation)**.** *Here $E_t$ is not $\bar{E}_t$ because of perfect information rational expectation*

**Second way to acquire NKPC**
We need to know two methods of price-setting:

- Price adjusting in certainty, Taylor, 1978[7]
- price adjusting in randomness, Calvo, 1983[8]

**Note 22** (Price adjusting in certainty vs in randomness)**.**

**Taylor**
a, b and c denotes three firms in a monopolistic competitive market. they adjust price by turns and one person per period.
Q1: how does one choose price
$P_A^{\triangle} = \frac{}{3}$, the upscript $\triangle$ stands for staying put in the next three periods.

[6] See Mankiw and Reis 2002 for more details of acceptable price-setting equation
[7] Taylor, John B. "Staggered Wage Setting in a Macro Model." The American Economic Review 69, no. 2 (1979): 108–13.
[8] Calvo, Guillermo A., 1983. "Staggered prices in a utility-maximizing framework," Journal of Monetary Economics, Elsevier, vol. 12(3), pages 383-398, September.

Q2: what is the total price
$P_t$ =
**Calvo**
each firm adjust price at probablity $\theta^t$ at period $t$.

Now sticky price price-adjusting equation

$$\hat{p}_t^{\triangle} = \underbrace{(1-\theta\beta)\sum_{k=0}^{\infty}(\theta\beta)^k E_t P_{t+k}^*}_{\text{weighted average}} \tag{402}$$

$$\Rightarrow \hat{p}_t^{\triangle} = (1-\theta\beta)\hat{p}_t^* + \theta\beta E_t \hat{p}_{t+1}^{\triangle} \text{ (sticky PAE)} \tag{403}$$

The $\theta$ is the opposite of the one in previous PAE, here rather is to stay put. price adjusting based on acceptable price setting equation in future periods, $\beta$ mean subjective discount.
With the expression of total price being

$$\Rightarrow \hat{p}_t = (1-\theta)\sum_{k=0}^{\infty}\theta^k \hat{p}_{t-k}^{\triangle} \tag{404}$$

$$\Rightarrow \hat{p}_t = (1-\theta)\hat{p}_t^{\triangle} + \theta\hat{p}_{t-1} \text{ (expression of total price based on APSE and PAE)} \tag{405}$$

Based on Acceptable Price-setting Equation (from monopolistic competition) and Total Price (from sticky price), one can quickly conduct NKPC.

**Note 23** (prices in DNK)**.**

starred price means acceptable price, triangled price means adjusting price

From the first equation $\hat{p}_t = (1-\theta)\sum_{k=0}^{\infty}\theta^k \hat{p}_{t-k}^{\triangle}$ we have

$$\hat{p}_t = (1-\theta)P_t^{\triangle} + \theta P_{t-1} \tag{406}$$

$$\Rightarrow (1-\theta)P_t^{\triangle} = \hat{p}_t - \theta\hat{p}_{t-1} \tag{407}$$

$$\Rightarrow (1-\theta)E_t P_{t+1}^{\triangle} = E_t\hat{p}_{t+1} - \theta\hat{p}_t \tag{408}$$

$$\Rightarrow (1-\theta)\beta E_t \hat{P}_{t+1}^{\triangle} = \beta E_t \hat{p}_{t+1} - \theta\beta\hat{p}_t \tag{409}$$

From the second equation $\hat{p}_t = (1-\theta)\hat{p}_t^{\triangle} + \theta\hat{p}_{t-1}$, first use sticky price price-adjusting equation $\hat{p}_t^{\triangle} = (1-\theta\beta)\hat{p}_t^* + \theta\beta E_t\hat{p}_{t+1}^{\triangle}$ to substitute

$$\Rightarrow \hat{p}_t - \theta\hat{p}_{t-1} = (1-\theta)\hat{p}_t^{\triangle} = (1-\theta)(1-\theta\beta)\hat{p}_t^* + (1-\theta)\theta\beta E_t P_{t+1}^{\triangle} \tag{410}$$

then use APSE $\hat{p}_t^* = \hat{p}_t + \gamma\tilde{y}_t$ to substitute

$$\Rightarrow \hat{p}_t - \theta\hat{p}_{t-1} = (1-\theta)(1-\theta\beta)(\hat{p}_t + \gamma\tilde{y}_t) + (1-\theta)\theta\beta E_t P_{t+1}^{\triangle} \tag{411}$$

use the conclusion $(1-\theta)E_t P_{t+1}^{\triangle} = E_t\hat{p}_{t+1} - \theta\hat{p}_t$ above

$$\Rightarrow \hat{p}_t - \theta\hat{p}_{t-1} = (1-\theta)(1-\theta\beta)(\hat{p}_t + \gamma\tilde{y}_t) + \theta\beta E_t\hat{p}_{t+1} - \theta^2\beta\hat{p}_t \tag{412}$$

$$\Rightarrow [1+\theta^2\beta - (1-\theta)(1-\theta\beta)]\hat{p}_t - \theta P_{t-1} = (1-\theta)(1-\theta\beta)\gamma\tilde{y}_t + \theta\beta E_t \text{ (break APSE)} \tag{413}$$

$$\Rightarrow [1+\theta^2\beta - (1-\theta)(1-\theta\beta)]\hat{p}_t - \theta P_{t-1} - \theta\beta\hat{p}_t = (1-\theta)(1-\theta\beta)\gamma\tilde{y}_t + \theta\beta(E_t\hat{p}_{t+1} - \hat{p}_t) \tag{414}$$

$$\Rightarrow \theta(\hat{p}_t - \hat{p}_{t-1}) = (1-\theta)(1-\theta\beta)\gamma\tilde{y}_t + \theta\beta(E_t\hat{P}_{t+1} - \hat{p}_t) \text{ (match terms so there is inflation)} \tag{415}$$

$$\Rightarrow \theta\hat{\pi}_t = (1-\theta)(1-\theta\beta)\gamma\tilde{y}_t + \theta\beta E_t\hat{\pi}_{t+1} \text{ (here comes inflation)} \tag{416}$$

$$\Rightarrow \hat{\pi}_t = k\hat{y}_t + \beta E_t\hat{\pi}_{t+1} \text{ (NKPC)} \tag{417}$$

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### 8.2.3 Dynamic IS curve

Recalling that household

$$\hat{y}_t = E_t\hat{y}_{t+1} - \frac{1}{\sigma}(\hat{i}_t - E_t\hat{\pi}_{t+1}) \tag{418}$$

$$\Rightarrow \hat{y}_t^f = E_t\hat{y}_{t+1} - \frac{1}{\sigma}(\hat{i}_t - 0) \text{ (inflation elastic form is assumed as 0)} \tag{419}$$

$$\Rightarrow \tilde{y}_t = E_t\tilde{y}_{t+1} - \frac{1}{\sigma}(\hat{i}_t - E_t\hat{\pi}_{t+1}) \text{ (dynamic IS curve, DIS)} \tag{420}$$

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## 8.3 Equilibrium of model

Now NKPC and DIS each represents supply and demand. The two equations have three unkown variables $\hat{\pi}_t, \tilde{y}_t, \hat{i}_t$, applying the model does not have a solution. It is so that we need to add another variable into the model. There are two ways making DNK model solvable, if not monetary supply rule then Taylor rule.

### 8.3.1 DNK with money supply rule

$$\text{DNK} \begin{cases} \hat{\pi}_t = \beta E_t\hat{\pi}_{t+1} + k\tilde{y}_t \text{ (NKPC)} \\ \tilde{y}_t = E_t\tilde{y}_{t+1} - \frac{1}{\sigma}(\hat{i}_t - E_t\hat{\pi}_{t+1}) \text{ (DIS)} \\ \hat{m}_t = \hat{p}_t + \hat{y}_t - \eta\hat{i}_t \text{ (Money supply rule, used to be ad hoc in RBC)} \end{cases} \tag{421}$$

Seems now three equations have even more unkown variables $\hat{\pi}_t, \tilde{y}_t, \hat{i}_t, \hat{p}_t, m_t$.
But hat $\pi_t, \hat{p}_t$ are the same thing because

$$\hat{\pi}_t = \hat{p}_t - \hat{p}_{t-1} \tag{422}$$

$$\Rightarrow \hat{p}_t = \hat{\pi}_t + \hat{p}_{t-1} \tag{423}$$

$$\Rightarrow \hat{m}_t = \hat{\epsilon}_t \tag{424}$$

this transfer works either way.
Let the $m_t$ money supply be exogenous for simplicity.

**Note 24** (Exogenous variables in DNK model)**.** *Only technology and money supply are considered real exogenous variables here.*

So we are left with three equations and three unkowns, the model is now solutionable. The final result will be a function of $m_t$.

### 8.3.2 DNK with interest rate rule

$$\text{DNK}\begin{cases}\hat{\pi}_t = \beta E_t\hat{\pi}_{t+1} + k\tilde{y}_t \text{ (NKPC)}\\ \tilde{y}_t = E_t\tilde{y}_{t+1} - \frac{1}{\sigma}(\hat{i}_t - E_t\hat{\pi}_{t+1}) \text{ (DIS)}\\ \hat{i}_t = \phi_\pi\hat{\pi}_t + \phi_y\hat{y}_t + \nu_t \text{ (Interest rate rule)}\end{cases} \tag{425}$$

$$\text{where } \nu_t = \rho_v\nu_{t-1} + \epsilon_t^\nu \text{ is the exogenous shock of interest rate, with the feature of AR1} \tag{426}$$

**Note 25** (Interest rate rule)**.**

$$\hat{i}_t = \phi_\pi \cdot (\hat{\pi}_t - \hat{\pi}^*) + \phi_y \cdot (\hat{y}_t - y_t^f) + \nu_t \tag{427}$$

If the target parameters $\hat{\pi}^*$ and $y_t^f$ are given, it is Taylor Rule.
Operation parameters $\phi_\pi$ and $\phi_y$ are decided based on past experience.

**Acquiring equilibrium and prove it is the only equilibrium**
Rewrite the DIS

$$\Rightarrow \tilde{y}_t = E_t\tilde{y}_{t+1} - \frac{1}{\sigma}[(\phi_\pi\hat{\pi}_t + \phi_y\hat{y}_t + \nu_t) - E_t\hat{\pi}_{t+1}] \text{ (apply interest rate rule to the second equation)} \tag{428}$$

$$= E_t\tilde{y}_{t+1} - \frac{1}{\sigma}\{[\phi_\pi \cdot (\beta E_t\hat{\pi}_{t+1} + k\tilde{y}_t) + \phi_y\hat{y}_t + \nu_t] - E_t\hat{\pi}_{t+1}\} \text{ (put the first in)} \tag{429}$$

$$= E_t\tilde{y}_{t+1} + \frac{1-\beta\phi_\pi}{\sigma}E_t\pi_{t+1} - \frac{\phi_\pi k + \phi_y}{\sigma}\tilde{y}_t - \frac{\nu_t}{\sigma} \tag{430}$$

$$= \frac{1}{\sigma + \phi_\pi k + \phi_y}[\sigma E_t\tilde{y}_{t+1} + (1-\beta\phi_\pi)E_t\hat{\pi}_{t+1} - \nu_t] \text{ (the relationship of } y_t \text{ and } y_{t+1} \text{ and } \pi_{t+1}) \tag{431}$$

Similarly we can change the NKPC

$$\hat{\pi}_t = \frac{1}{\sigma + \phi_\pi k + \phi_y}\{k\sigma E_t\tilde{y}_{t+1} + [k + \beta(\sigma + \phi_y)]E_t\hat{\pi}_{t+1} - \nu_t\} \tag{432}$$

Therefore three equations are two now.
Iteration in vector form

$$\begin{bmatrix}\tilde{y}_t\\ \hat{\pi}_t\end{bmatrix} = \frac{1}{\sigma + \phi_\pi k + \phi_y}\begin{bmatrix}\sigma & 1-\beta\phi_\pi\\ k\sigma & k + \beta(\sigma + \phi_y)\end{bmatrix}\begin{bmatrix}E_t\hat{y}_{t+1}\\ E_t\hat{\pi}_{t+1}\end{bmatrix} - \frac{1}{\sigma + \phi_\pi k + \phi_y}\begin{bmatrix}1\\ k\end{bmatrix}\nu_t \tag{433}$$

$$\Rightarrow \begin{bmatrix}\tilde{y}_t\\ \hat{\pi}_t\end{bmatrix} = A\begin{bmatrix}C_{t-1}\\ K_{t-1}\end{bmatrix} - B\nu_t \text{ (at RHS one period ahead meaning the whole equation is foresight-seeing)} \tag{434}$$

$$\Rightarrow \lambda^2 - \frac{\sigma + k + \beta(\sigma + \phi_y)}{\sigma + \phi_\pi k + \phi_y}\lambda + \frac{\sigma\beta}{\sigma + \phi_\pi k + \phi_y} = 0 \tag{435}$$

**Note 26** (Technique in Ramsey and DNK)**.**

From the perspective of techniques in persuiing solution, it is greatly similar to the way we did with Ramsey model, where the model is one period backward at RHS.

**Note 27** (HNKPC)**.** *Hybrid New Keynesian Philips Curve*

Since NKPC lacks persistence in practical use (shocks are too slow to effects), we might as well add a 1-period-lagged inflation to the RHS as adpative expectation. This is somehow a nominal shock since it gets into the DNK model through $m_t$.

$$\hat{\pi}_t = \beta E_t \hat{\pi}_{t+1} + k\tilde{y}_t \text{ (NKPC)} \tag{436}$$
$$\hat{\pi}_t = \hat{\pi}_{t-1} + \beta E_t \hat{\pi}_{t+1} + k\tilde{y}_t \text{ (HNKPC)} \tag{437}$$

$a_t$ is described as real shocks of DNK model. But it shows up too fast.
Later the concept of sticky information is added to the model to solve the problem.

**Note 28** (BK conditions)**.**

First brought up by Blanchard and Kahn, 1980

[9]

**Note 29.**

Lasalle, 1986

$$x^2 + bx + c = 0 \tag{438}$$
$$x_1, x_2 < 1 \iff \begin{cases} |c| < 1 \\ |b| < 1 + c \end{cases} \tag{439}$$

If the equilibrium exists and it is the only one, then

$$\begin{cases} |\frac{\sigma\beta}{\sigma+\phi_\pi k+\phi_y}| < 1 \Rightarrow \sigma(\beta-1) < 0 < \phi_\pi k + \phi_y \\ |\frac{\sigma+k+\beta(\sigma+\phi_y)}{\sigma+\phi_\pi k+\phi_y}| < 1 + \frac{\sigma\beta}{\sigma+\phi_\pi k+\phi_y} \Rightarrow k(\phi_\pi - 1) + (1-\beta)\phi_y > 0 \end{cases} \tag{440}$$

Method of Undetermined Coefficients:
Let's assume

$$\begin{cases} \tilde{y}_t = \psi_{y\nu}\nu_t \\ \hat{\pi}_t = \psi_{\pi\nu}\nu_t \end{cases} \tag{441}$$

[9] Also see Nakajima 2007 for similar techniques.

recall that

$$\hat{\pi}_t = \beta E_t \hat{\pi}_{t+1} + k\hat{y}_t \text{ (expected inflation)} \tag{442}$$

$$\psi_{\pi\nu}\nu_t = \beta E_t(\psi_{\pi\nu}\nu_{t+1}) + k\hat{y}_t \text{ (iterating)} \tag{443}$$
$$= \beta E_t[\psi_{\pi\nu}(\rho_\nu \nu_t + \epsilon^{\nu}_{t+1})] + k\hat{y}_t \text{ (expand } \nu_{t+1} \text{ since it's AR1 shock of } i_t) \tag{444}$$
$$= \beta\psi_{\pi\nu}\rho_\nu\nu_t + k\psi_{y\nu}\nu_t \text{ (preassuming expectation of next period noise is } 0) \tag{445}$$
$$\Rightarrow (1-\beta\rho_\nu)\psi_{\pi\nu} = k\psi_{y\nu} \tag{446}$$
$$\Rightarrow k\psi_{y\nu} = \frac{1-\beta\rho_\nu}{k}\psi_{\pi\nu} \text{ (relationship of two undetermined coefficients)} \tag{447}$$

Using DIS

$$\tilde{y}_t = E_t\tilde{y}_{t+1} - \frac{1}{\sigma}[\phi_\pi\hat{\pi}_t + \phi_y\hat{y}_t + \nu_t) - E_t\hat{\pi}_{t+1}] \tag{448}$$

using interest rate rule

$$\Rightarrow \psi_{y\nu}\nu_t = \psi_{y\nu}E_t\nu_{t+1} - \frac{1}{\sigma}(\phi_\pi\psi_{\pi\nu}\nu_t + \phi_y\psi_{y\nu}\nu_t + \nu_t - \psi_{\pi\nu}E_t\nu_{t+1}) \tag{449}$$
$$= \psi_{y\nu}\rho_\nu\nu_t - \frac{1}{\sigma}(\phi_\pi\psi_{\pi\nu}\nu_t + \phi_y\psi_{y\nu}\nu_t + \nu_t - \psi_{\pi\nu}\rho_\nu\nu_t) \tag{450}$$
$$= \frac{\sigma\rho_\nu - \phi_y}{\sigma}\phi_{y\nu}\nu_t - \frac{\phi_\pi - \rho_\nu}{\sigma}\psi_{\pi\nu}\nu_t - \frac{1}{\sigma}\nu_t \tag{451}$$
$$\Rightarrow \sigma\psi_{y\nu} = (\sigma\rho_\nu - \phi_y)\psi_{y\nu} - (\phi_\pi - \rho_\nu)\psi_{\pi\nu} - 1 \text{ (elimination of } \nu_t) \tag{452}$$
$$\Rightarrow (\sigma - (\sigma\rho_\nu - \phi_y))\psi_{y\nu} + (\phi_\pi - \rho_\nu)\psi_{\pi\nu} = -1 \tag{453}$$
$$\Rightarrow \psi_{\pi\nu} = \frac{-k}{\cdots} = -k\wedge_\nu \tag{454}$$
$$\Rightarrow \psi_{y\nu} = -(1-\beta\rho_\nu)\wedge_\nu \tag{455}$$

## 8.4 Analyze exogenous shocks

Now policy change will reflect in the parameter $\epsilon$, making policy analysis possible.
we use $\nu$ to denote the exogenous shock
last section we have expression of $\nu$
$\Rightarrow$ to analyze the shock would mean fully using it.

**Note 30** (what is a shock?)**.**

In a model with uncertainty, variable has the feature of causing change in variation by change itself can cause a shock.
But it was later broadened as change of exogenous variable in a model without uncertainty.

recall output gap at equilibrium

$$\begin{aligned}
\hat{y}_t - \hat{y}_t^f = \hat{y}_t^* &= \psi_{y\nu}\nu_t && (456)\\
&= -(1-\beta\rho_\nu) \wedge_\nu \nu_t && (457)\\
&= -(1-\beta\rho_\nu) \wedge_\nu (\rho_\nu \nu_{t-1} + \epsilon_t^\nu) && (458)\\
&= -(1-\beta\rho_\nu) \wedge_\nu \rho_\nu\nu_{t-1} - (1-\beta\rho_\nu) \wedge_\nu \epsilon_t^\nu && (459)\\
&= \rho_\nu \tilde{y}_{t-1} - (1-\beta\rho_\nu) \wedge_\nu \epsilon_t^\nu \ (\tilde{y}_{t-1} \text{ backward one period}) && (460)
\end{aligned}$$

**Note 31.** *we can only do analysis at equil*

similiar with inflation at equilibrium

$$\begin{aligned}
\hat{\pi}_t^* &= \psi_{\pi\nu}\nu_t && (461)\\
&= -k \wedge_\nu \nu_t && (462)\\
&= -k \wedge_\nu (\rho_\nu \nu_{t-1} + \epsilon_t^\nu) && (463)\\
&= \rho_\nu \pi_{t-1} - k \wedge_\nu \epsilon_t^\nu && (464)
\end{aligned}$$

interest rate at equilibrium

$$\begin{aligned}
\hat{r}_t^* &= \hat{i}_t - E_t\hat{\pi}_{t+1} \text{ (Real interest rate at equilibrium)} && (465)\\
&= \sigma(E_t\tilde{y}_{t+1} - \hat{y}_t) \leftarrow \text{DIS} && (466)\\
&= \sigma(1-\beta\rho_\nu)(1-\rho_\nu) \wedge_\nu \nu_t && (467)\\
\Rightarrow \hat{i}_t^* &= \hat{r}_t + E_t\hat{\pi}_{t+1} \text{Nominal interest rate at equilibrium} && (468)\\
&= [\sigma(1-\beta\rho_\nu)(1-\rho_\nu) - k\rho_\nu] \wedge_\nu \nu_t && (469)
\end{aligned}$$

moreever we can do analysis with money supply using quantity equation of money

$$\begin{aligned}
\hat{m}_t^* &= \hat{p}_t^* + \hat{y}_t^* - \eta\hat{i}_t^* \text{ (LHS is solvable since all variables at RHS has been solved)} && (470)\\
&= \text{ (assuming } \hat{y}_t^{f*} = 0) && (471)\\
&= && (472)\\
&= \text{ use solutions from above} && (473)\\
&\Rightarrow && (474)\\
&= && (475)\\
\frac{d\hat{m}_t^*}{d\nu_t} &= && (476)\\
&= \text{showing the result is decided by mutiple joint forces} && (477)
\end{aligned}$$

**Note 32** (quantity equation of money)**.** $MV = Py$
*While this theory was originally formulated by Polish mathematician Nicolaus Copernicus in 1517, it was popularized later by economists Milton Friedman and Anna Schwartz after the publication of their book, "A Monetary History of the United States, 1867-1960," in 1963.*

### 8.5 Natural rates, output gaps, and model limits

*Planned section.* Distinguish the natural real interest rate, flexible-price output, and efficient output. Check which shocks and distortions enter each object before interpreting policy responses or welfare losses.

## 9 Sticky Wage Model

Before good market is price-sticky and monopolistic competitive, now the setting goes to labor market.

$$Y_{it} = (\frac{P_{it}}{P_t})^{-\epsilon_P} Y_t \text{ (good demand curve, } \epsilon \text{ is substitute elasticity)} \tag{478}$$

$$P_t = (\int_0^1 P_{it}^{1-\epsilon} di)^{\frac{1}{1-\epsilon_P}} \text{ (total price)} \tag{479}$$

$$W_t = W_{it} = \int_0^1 W_{it} di \tag{480}$$

$$P_{it}^* = c + \alpha \tilde{y}_t \text{ (desired price)} \tag{481}$$

labor market m.c. (as in classical model) and sticky wage (partial rigidity, in contrary to total rigidity where all wages are rigid).
1) labor demand curve and total wage price level

$$\begin{aligned} &\min \int_0^1 W_{jt} N_{ijt} dj \\ &s.t. A_t [(\int_0^1 N_{ijt}^{\frac{\epsilon_W - 1}{\epsilon_W}} dj)^{\frac{\epsilon_W}{\epsilon_W - 1}}]^{1-\alpha} \geqslant A_t N_{it}^{1-\alpha} \end{aligned} \tag{482}$$

where $i$ indexing firm, $j$ indexing labor, $t$ is time
Form a Lagrangian

$$\mathscr{L} = \int_0^1 W_{jt} N_{ijt} dj + W_t \{A_t [(\int_0^1 N_{ijt}^{\frac{\epsilon_w - 1}{\epsilon_w}} dj)^{\frac{\epsilon_W}{\epsilon_W - 1}}]^{1-\alpha} - A_t N_{it}^{1-\alpha}\} \tag{483}$$

where Lagrangian multiplier is total wage level
Solving the optimization problem results

$$\Rightarrow \begin{cases} N_{ijt} = (\frac{W_{jt}}{W_t})^{-\epsilon_W} N_{it} \text{ (labor supply equation)} \\ W_t = (\int_0^1 W_{jt}^{1-\epsilon_W} dj)^{\frac{1}{1-\epsilon_W}} \text{ (expression of total wage level)} \end{cases} \tag{484}$$

2) UMP of household
in utility function N is labor demand instead of labor supply as before. this is because of m.c. labor market.

$$\begin{aligned} u &= u(C_t, N_t) \\ &= u(C_t, \{N_{jt}\}) \\ &= \frac{C_t^{1-\sigma} - 1}{1-\sigma} - \int_0^1 \frac{N_j^{1+\varphi}}{1+\varphi} dj \text{ (CES style)} \end{aligned} \tag{485}$$

the UMP can be rewritten as

$$\begin{aligned}\max_{\{C_t,W_t^*\}}\sum_{t-0}^{\infty}\beta^t u(C_t,\int_0^1 N_{jt}dj) \iff \max\sum_{t=0}^{\infty}\beta^t u(C_t,\int_0^1(\frac{W_t^*}{W_t})^{-\epsilon_W}N_{it}dj)\\ s.t. P_tC_t+Q_tB_t\leqslant\int_0^1 W_t^*(\frac{W_t^*}{W_t})^{-\epsilon_W}N_{it}dj+B_{t-1}\end{aligned} \tag{486}$$

here $W_t^*$ instead of original because all wages can be adjusted, eventually they will be the same
a simplified version would be

$$\begin{aligned}&\max\sum_{t=0}^{\infty}\beta^t u(C_t,(\frac{W_t^*}{W_t})^{-\epsilon_W}N_{it})\\ &s.t. P_tC_t+Q_tB_t\leqslant W_t^*(\frac{W_t^*}{W_t})^{-\epsilon_W}+B_{t-1}-T_t\end{aligned} \tag{487}$$

$$\begin{aligned}&\max_{W_0^*}\sum_{t=0}^{\infty}\beta^t\theta_W^t u(C_{t|0},N_{t|0})\\ &s.t.\begin{cases}C_{t|0}=\frac{1}{P_t}[W_0^*(\frac{W_0^*}{W_t})^{-\epsilon_W}N_{it}+B_{t-1|0}-T_{t|0}-Q_{t-1|t}B_{t|0}]\\ N_{t|0}=(\frac{W_0^*}{W_t})^{-\epsilon_W}N_{it}\text{ (with labor demand curve, left with only }W_0^*)\end{cases}\end{aligned} \tag{488}$$

sticky wage, where $\theta\equiv Prob(\text{stay put})$. $\theta$ is before utility function because the whole equation is based on period 0, entailing a prob for the whole equation to stay put.
put two constraints in the max part, now the optimization problem is a unconstrainted one

$$\max_{W_0^*}\sum_{t=0}^{\infty}(\beta\theta_W)^t u(\frac{1}{P_t}[W_0^*(\frac{W_0^*}{W_t})^{-\epsilon_W}N_{it}+B_{t-1|0}-T_{t|0}-Q_{t-1|t}B_{t|0}],(\frac{W_0^*}{W_t})^{-\epsilon_W}N_{it}) \tag{489}$$

$$FOC\Rightarrow E_0\sum_{t=0}^{\infty}(\beta\theta_W)^t[u_C(C_{t|0},N_{t|0})\frac{1}{P_t}(1-\epsilon_W)(\frac{W_0^*}{W_t})^{-\epsilon_W}N_{it}-u_N(C_{t|0},N_{t|0})\epsilon_W(\frac{W_0^*}{W_t})^{-(\epsilon_W)^{-1}}\frac{1}{W_t}N_{it})] \tag{490}$$

$$=E_0\sum_{t=0}^{\infty}(\beta\theta_W)^t[u_C(C_{t|0},N_{t|0})\frac{1}{P_t}(1-\epsilon_W)(\frac{W_0^*}{W_t})^{-\epsilon_W}W_t^{\epsilon_W}N_{it}-u_N(C_{t|0},N_{t|0})\epsilon_W(\frac{W_0^*}{W_t})^{-(\epsilon_W)}W_t^{\epsilon_W}N_{it}\frac{1}{W_0^*})] \tag{491}$$

move some exogenous variables out since they basically can be treated as constants

$$\Rightarrow(1-\epsilon_W)(W_0^*)^{-\epsilon_W}\dots \tag{492}$$

$$=\epsilon_W(W_0^*)^{-\epsilon_W^{-1}} \tag{493}$$

$$\Rightarrow W_0^*=\frac{\epsilon_W}{1-\epsilon_W}\frac{E_0\sum_{t=0}^{\infty}(\beta\theta_W)^t u_N(C_{t|0},N_{t|0})W_t^{\epsilon_W}N_{it}}{E_0\sum_{t=0}^{\infty}(\beta\theta_W)^t u_C(C_{t|0},N_{t|0})\frac{1}{P_t}W_t^{\epsilon_W}N_{it}} \tag{494}$$

use marginal substitute ratio, where $MRS_{t|0} \equiv -\frac{U_N(C_{t|0})}{U_C(C_{t|0})}$, the equation is essentially an inflation equation

$$W_0^* = \frac{\epsilon_W}{\epsilon_W - 1} \frac{E_0 \sum_{t=0}^{\infty} (\beta\theta_W)^t u_N(C_{t|0}, N_{t|0}) MRS_{t|0} W_t^{\epsilon_W} N_{it}}{E_0 \sum_{t=0}^{\infty} (\beta\theta_W)^t u_C(C_{t|0}, N_{t|0}) \frac{1}{P_t} W_t^{\epsilon_W} N_{it}} \tag{495}$$

$$\Rightarrow \frac{W_0^*}{P_0} = \frac{\epsilon_W}{\epsilon_W - 1} MRS_{t|0} \text{ (if elastic wage and m.c., there will be no equation above)} \tag{496}$$

now based on the perspective at period $t$

$$W_t^* = \frac{\epsilon_W}{\epsilon_W - 1} \frac{E_t \sum_{k=0}^{\infty} (\beta\theta_W)^k u_C(C_{t+k|t}, N_{t+k|t}) MRS_{t+k|t} W_{t+k|t}^{\epsilon_W}}{E_t \sum_{k=0}^{\infty} (\beta\theta_W)^k u_C(C_{t+k|t}, N_{t+k|t}) \frac{1}{P_{t+k}} W_{t+k|t}^{\epsilon_W}} \text{ (at t period instead 0 period)} \tag{497}$$

$$\Rightarrow \frac{W_t^*}{P_t} = \frac{\epsilon_W}{\epsilon_W - 1} MRS \text{ (if elasticity wage and monopolistic competitive at t period)} \tag{498}$$

**Note 33** (M.C. firm choice variable for optimization problem)**.** *MC firm can choose price max or quantity max*

# 10 Labor Markets and Unemployment

*Planned chapter.*

## 10.1 Employment, unemployment, and participation

Define the margins hidden by a single aggregate labor-supply variable. Distinguish changes in hours from movements into and out of employment.

## 10.2 Search, matching, and wage determination

Introduce a basic matching framework to explain vacancies and unemployment. State how wage setting and hiring incentives determine job creation.

## 10.3 Adjustment and persistence

Compare search frictions with nominal wage rigidity. Examine which mechanisms can make employment recover slowly and connect a recession to longer-run output.

# Part III
# Open Economy

An open economy can trade goods and claims on future income. This changes the connection between domestic saving and investment, introduces relative prices between countries, and allows shocks to cross borders. We extend the earlier models before comparing policy choices in Part IV.

# 11 International Accounts and Intertemporal Trade

*Planned chapter.*

## 11.1 Trade balances, current accounts, and external wealth

Derive the links between national saving, investment, and the current account. Distinguish transaction flows from valuation changes in the international investment position.

## 11.2 Consumption smoothing and international borrowing

Start with a small open endowment economy. Derive its intertemporal budget constraint and explain how expected income affects the current account.

## 11.3 Investment and external adjustment

Add capital accumulation and examine how borrowing finances investment. State borrowing and solvency conditions before discussing external adjustment.

# 12 Trade and Relative Prices

*Planned chapter.*

## 12.1 Comparative advantage, trade gains, and trade costs

Use a simple trade model to explain specialization and gains from exchange. Introduce tariffs and other trade costs to connect trade policy to allocation.

## 12.2 Terms of trade and the real exchange rate

Define both relative prices with explicit conventions. Trace how their changes affect purchasing power, production, and external balances.

## 12.3 Tradable goods, nontradable goods, and adjustment

Introduce the distinction between tradable and nontradable production. Explain how expenditure and resources move between sectors during adjustment.

# 13 Open Economy Fluctuations and Exchange Rates

*Planned chapter.*

## 13.1 The small open economy RBC model

Extend the real benchmark to international borrowing and investment. Compare productivity, world-interest-rate, and terms-of-trade shocks.

## 13.2 Capital flows, interest parity, and risk

Derive the relevant asset-pricing conditions and distinguish covered parity from an uncovered expected-return condition. Introduce risk premia and foreign-currency balance-sheet exposure.

## 13.3 Nominal rigidities and exchange-rate regimes

Introduce an open economy New Keynesian benchmark. Compare the adjustment mechanisms under fixed and flexible exchange rates, and state the constraints created by capital mobility.

# Part IV
# Money, Finance, and Macroeconomic Policy

The earlier parts establish the real and nominal benchmarks. We now add financial constraints and differences between households, then compare the transmission and welfare effects of monetary and fiscal policy. Open economy mechanisms remain part of that comparison.

# 14 Money, Banking, and Central Bank Operations

*Planned chapter.*

## 14.1 Money, payments, and financial claims

Distinguish money from other assets and identify the services that give it a role in household and firm decisions.

## 14.2 Banks and balance sheets

Set out the balance sheets of households, commercial banks, and the central bank. Explain lending, deposits, reserves, and the limits to credit creation.

## 14.3 Implementing monetary policy

Connect the model's policy rate to the central bank's operating framework. Distinguish reserve provision, asset purchases, and changes in borrowing costs.

# 15 Financial Frictions and Aggregate Adjustment

*Planned chapter.*

## 15.1 Borrowing limits and collateral

Derive a simple borrowing constraint and show how it changes optimal choices. Connect collateral values to spending and investment.

## 15.2 Intermediation and financial amplification

Introduce a constraint on intermediary balance sheets. Explain how losses, leverage, and credit spreads can amplify a real or financial disturbance.

## 15.3 Financial distress and policy intervention

Distinguish liquidity problems from insolvency. State the mechanisms and trade-offs behind liquidity support, restructuring, and macroprudential policy.

# 16 Household Heterogeneity and Policy Transmission

*Planned chapter.*

## 16.1 Income risk, insurance, and precautionary saving

Introduce uninsurable income risk and borrowing limits. Explain how the distribution of liquid wealth affects saving and consumption.

## 16.2 Marginal propensities to consume and redistribution

Compare households' spending responses to income and wealth changes. Trace how taxes, transfers, and interest payments redistribute resources.

## 16.3 From representative to heterogeneous households

Use a two-agent benchmark before motivating a richer distribution of households. Identify which policy questions require the additional heterogeneity.

# 17 Monetary Policy

This chapter extends the interest-rate rules used to close the New Keynesian model into a systematic policy comparison. The existing welfare derivation is retained below; the preceding sections are outlines.

## 17.1 Transmission and policy shocks

*Planned section.* Separate a systematic policy response from an unexpected policy disturbance. Compare intertemporal, credit, redistribution, and exchange-rate channels.

## 17.2 Rules, commitment, and discretion

*Planned section.* Compare equilibrium determination with welfare-based policy design. Explain how credibility and promises about future policy affect current choices.

## 17.3 The interest-rate lower bound and balance-sheet policies

*Planned section.* State when the policy-rate constraint binds and which transmission channels remain. Examine forward guidance and asset purchases under explicit assumptions.

## 17.4 Optimal monetary policy

A significant paper Mankiw and Reis 2002 used a new keynesian macroeconomic model to acquire the optimal monetary policy. [10]

**Note 34** (why is there welfare loss?)**.**

monopoly and sticky price cause deviation from Pareto optimality.

**Note 35** (Optimal monetary policy)**.**

No concept of optimality can be found in policy analysis in the perspective of household or firm. But the government can use welfare loss fucntion in its optimality to maximize social welfare. In the process of achieving maximized social welfare, the monetary policy is the so-called optimal monetary policy.

To acquire the optimal monetary policy, the welfare loss function must be previously introduced.
To acquire WLF, we use second order taylor expansion
first order Taylor expansion

$$\frac{X_t - X}{X} = \ln X_t - \ln X = x_t - x = \hat{x}_t \tag{499}$$

[10] N. Gregory Mankiw, Ricardo Reis, Sticky Information versus Sticky Prices: A Proposal to Replace the New Keynesian Phillips Curve, The Quarterly Journal of Economics, Volume 117, Issue 4, November 2002, Pages 1295–1328, https://doi.org/10.1162/003355302320935034

second order Taylor expansion

$$\begin{aligned}\frac{X_t - X}{X} &= \frac{1}{X}e^{x_t} - 1 \\ &= (\frac{1}{X}e^{x} - 1) + \frac{1}{X}e^{x}(x_t - x) + \frac{1}{2}\frac{1}{X}e^{x}(x_t - x)^2 \\ &= \hat{x}_t + \frac{1}{2}\hat{x}_t^2\end{aligned} \tag{500}$$

Similarly

$$\Rightarrow \begin{cases} \frac{C_t - C}{C} \approx \hat{c}_t + \frac{1}{2}\hat{c}_t^2 \\ \frac{N_t - N}{N} \approx \hat{n}_t + \frac{1}{2}\hat{n}_t^2 \end{cases} \tag{501}$$

Bi-variate 2o taylor expansion for a utility function where people gain utilities from consuming goods and enjoying leisure.

$$u(C_t, N_t) \approx u(C, N) + u_C^C(\frac{C_t - C}{C}) + u_N^N(\frac{N_t - N}{N}) + \frac{1}{2}u_{CC}^{C^2}\frac{(C_t - C)^2}{C} + \frac{1}{2}u_{NN}^{N^2}\frac{(N_t - N)^2}{N} \tag{502}$$

$$= u(C, N) + u_C C(\hat{c}_t + \frac{1}{2}\hat{c}_t^2) + u_N N(\hat{n}_t + \frac{1}{2}\hat{n}_t^2) + \frac{1}{2}u_{CC}C^2(\hat{c}_t + \frac{1}{2}\hat{c}_t^2) + \frac{1}{2}u_{NN}N^2(\hat{n}_t + \frac{1}{2}\hat{n}_t^2) \tag{503}$$

$$= u(C, N) + u_C C(\hat{c}_t + \frac{1}{2}\hat{c}_t^2) + u_N N(\hat{n}_t + \frac{1}{2}\hat{n}_t^2) + \frac{1}{2}u_{CC}C^2\hat{c}_t + \frac{1}{2}u_{NN}N^2\hat{n}_t \text{ (reservate only to 2o)} \tag{504}$$

$$= u(C, N) + u_C C(\hat{c}_t + \frac{1 + \frac{u_{CC}}{u_C}C}{2}\hat{c}_t) + u_N N(\hat{n}_t + \frac{1 + \frac{u_{NN}}{u_N}N}{2}\hat{n}_t) \tag{505}$$

$$\Rightarrow \frac{u(C_t, N_t) - u(C, N)}{u_C C} = (\hat{c}_t + \frac{1 + \frac{u_{CC}}{u_C}C}{2}\hat{c}_t^2) + \frac{u_N N}{u_C C}(\hat{n}_t + \frac{1 + \frac{u_{NN}}{u_N}}{2}\hat{n}_t^2) \tag{506}$$

$$\because \hat{c}_t = \hat{y}_t \text{ (market clearing at equilibrium)} \tag{507}$$

$$u(C_t, N_t) = \frac{C_t^{1-\sigma} - 1}{1 - \sigma} - \frac{N_t^{1+\varphi} - 1}{1 + \varphi} \iff \begin{cases} -\frac{u_{CC}}{u_C}C = \sigma \\ -\frac{u_{NN}}{u_N}N = \varphi \end{cases} \tag{508}$$

**Note 36** (Coefficients of CES function)**.**

$\sigma$ is intertemporal substitute elasticy, $\varphi$ is frisch elasticity

continue to the last taylor expanded equation, introducing $var_i$ as the deviation of each firm's price

$$\ldots = (\hat{y}_t + \frac{1 - \sigma}{2}\hat{y}_t^2) + \frac{u_N N}{u_C C}\{\frac{1}{\alpha}(\hat{y}_t - a_t) + A \cdot var_i P_{it}\} + \frac{1 + \varphi}{2}[\frac{1}{\alpha}(\hat{y}_t - a_t) + A \cdot var_i P_{it}]^2 \tag{509}$$

$$= (\hat{y}_t + \frac{1 - \sigma}{2}\hat{y}_t^2) + \frac{u_N N}{u_C C}\frac{1}{\alpha}\{[(\hat{y}_t - a) + B \cdot var_i P_{it}] + \frac{1 + \varphi}{2\alpha}(\hat{y}_t - a_t)^2\}, B \equiv \alpha A \tag{510}$$

, here let's ignore many fluctuation terms from taylor expansion as they are extremely small and irrelevant to monetary policy.

$$-\frac{U_N}{U_C} = MP \cdot N = \alpha A N^{\alpha-1} \tag{511}$$

$$= \alpha \frac{C}{N} \Rightarrow \frac{u_N N}{u_C C} = -\alpha \text{ (can use this to simplify equation before)} \tag{512}$$

go back to before

$$\Rightarrow \ldots = (\hat{y}_t - \frac{1-\sigma}{2}\hat{y}_t^2) - \{(\hat{y}_t - a_t) + B \cdot var_i P_{it} + \frac{1+\varphi}{2\alpha}(\hat{y}_t - a_t)^2\} \tag{513}$$

$$= \frac{1-\sigma}{2}\hat{y}_t^2 - B \cdot var_i P_{it} - \frac{1+\varphi}{2\alpha}(\hat{y}_t - a_t)^2 \tag{514}$$

$$= \frac{1-\sigma}{2}\hat{y}_t^2 - B \cdot var_i P_{it} - \frac{1+\varphi}{2\alpha}(\hat{y}_t^2 - 2\hat{y}_t a_t + a_t^2) \tag{515}$$

$$= (\frac{1-\sigma}{2} + \frac{1+\varphi}{2\alpha})\hat{y}_t^2 + \frac{\sigma+\varphi}{\alpha}\hat{y}_t a_t - B \cdot var_i P_{it} \tag{516}$$

$$= \frac{\alpha(1-\alpha)+1+\varphi}{2\alpha}\hat{y}_t^2 + \frac{\sigma+\varphi}{\alpha}\hat{y}_t \hat{y}_t^f - B \cdot var_i P_{it} \text{ (remember when we work out the } \hat{y}_t^f\text{)} \tag{517}$$

$$= \underbrace{-\frac{1}{2}[\underbrace{C(\hat{y}_t - \hat{y}_t^f)}_{\text{deviation from output}} + \underbrace{D(\hat{\pi}_t - \pi^*)^2}_{\text{inflation}}]}_{\text{welfare loss}} \text{ (welfare loss function)} \tag{518}$$

the goal is to minimize the whole term

> **Note 37** (paradoxy in welfare loss function)**.**
>
> the two goals are contradicated according to philips curve

# 18 Fiscal Policy

*Planned chapter.*

## 18.1 Government budgets, debt, and Ricardian equivalence

Derive the government flow and intertemporal budget constraints. Explain the conditions for Ricardian equivalence and why they can fail.

## 18.2 Taxes, purchases, transfers, and public investment

Separate income, substitution, and resource effects. Compare the objectives and transmission mechanisms of different fiscal instruments.

## 18.3 Fiscal multipliers and financing

Define the multiplier and its horizon. Compare spending responses across financing arrangements, monetary policy rules, household structures, and open economy environments.

# 19 Policy Interaction and Constraints

*Planned chapter.*

## 19.1 Fiscal and monetary regimes

Study how the fiscal backing of public liabilities interacts with interest-rate policy, inflation, and equilibrium determination.

## 19.2 Debt sustainability and policy space

Derive debt-to-output dynamics and distinguish arithmetic from an assessment of sustainability. Include growth, interest rates, primary balances, and risk.

## 19.3 Policy combinations in an open economy

Bring together exchange-rate regimes, capital flows, domestic balance sheets, and international spillovers. Compare policy combinations under stated institutional and financial constraints.

# Part V
# Topics

The topics combine mechanisms from the earlier parts. Each begins with dated facts and measurement choices, compares competing explanations, and asks what evidence would distinguish them. Policy conclusions remain conditional on that evidence.

# 20 The Dollar in the International Monetary System

*Planned chapter.*

## 20.1 History and measures of international currency use

Separate reserve holdings, invoicing, payments, funding, and exchange-rate anchors. Define what each measure can establish about dollar dominance.

## 20.2 Network effects, invoicing, and safe assets

Explain the mechanisms that can sustain an international currency's use. Connect dollar pricing and demand for safe assets to the open economy models.

## 20.3 Benefits, obligations, and the international balance sheet

Distinguish seigniorage, financing advantages, and valuation effects. Examine the obligations and constraints associated with supplying safe and liquid international liabilities.

## 20.4 Global funding and policy spillovers

Trace dollar borrowing, funding stress, and the international transmission of US policy. Examine liquidity facilities and the role of financial sanctions.

## 20.5 Persistence, alternatives, and testable changes

Compare explanations of persistence with possible sources of change. Assess claims about de-dollarization separately across the measures defined at the start of the chapter.

# 21 China's Growth and Adjustment after the Pandemic

*Planned chapter.*

## 21.1 The baseline, timeline, and measurement

Separate the pre-pandemic baseline, pandemic years, and reopening period. Distinguish output levels from growth rates and real activity from nominal income. An observation after the pandemic does not identify a pandemic effect.

## 21.2 Demand, consumption, and employment

Compare cyclical explanations using household income, consumption, prices, and employment. State what each explanation predicts before assessing it.

## 21.3 Property, balance sheets, and local public finance

Map the links between housing, household and firm balance sheets, banks, and local government finances. Compare expenditure adjustment and debt resolution mechanisms.

## 21.4 Population, productivity, and resource allocation

Use growth accounting and the growth models to distinguish demographic, investment, and productivity explanations. Examine the allocation of resources across sectors and firms.

## 21.5 External demand and policy alternatives

Connect domestic adjustment to trade and international conditions. Compare policy alternatives through their mechanisms, constraints, distributional effects, and testable implications.